

# recon

## User Manual

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Repository · <https://github.com/thomas-starweb/recon>

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# About this manual

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recon is a versatile network reconnaissance CLI written in Rust. It started as a curl clone and grew into a multi-protocol investigation tool covering HTTP(S), TLS certificate inspection, DNS, WHOIS, ping, traceroute, barcode encoding and decoding, file compression and archiving, markdown/HTML/PDF conversion, and a full Rhai script engine that exposes every protocol probe and helper for automation.

This manual covers every user-facing flag, every script binding, and shows how to combine them. The built-in `recon --help <topic>` command remains the quick reference; this document is the long-form companion.

A PDF rendering of this manual lives alongside it at [MANUAL.pdf](#). It is regenerated every time the markdown changes — see `CLAUDE.md` for the maintenance policy. Generate locally with:

```
recon --md-to-pdf docs/MANUAL.md \  
  --toc --toc-depth 3 --gfm --unsafe-html --page-break-on-h1 \  
  --doc-title 'recon User Manual' \  
  -o docs/MANUAL.pdf
```

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The table of contents above is auto-generated from H1/H2/H3 headings. Every entry is a clickable anchor in the PDF. For a narrative walkthrough of the structure, the four parts are:

- **Part I — Getting started:** introduction, installation, quick start.
  - **Part II — CLI reference:** every flag grouped by area, with examples.
  - **Part III — Script engine:** the Rhai interpreter, every binding, many examples.
  - **Part IV — Appendices:** exit codes, env vars, config, glossary.
-

# Introduction

---

recon is a single-binary command-line tool. Its design goals, in rough order:

1. **Curl-compatible** where it overlaps with curl. Flag names, short forms, and behaviours match curl for HTTP(S) requests so existing tooling keeps working. `recon --help curl` lists the compatibility status.
  2. **Pure Rust** where possible. request + rustls for HTTP(S); hickory for DNS; comrak for markdown; flate2/brotli/zstd for compression; rxing for barcodes; tungstenite for WebSockets; rumqttn for MQTT. Where a pure-Rust path doesn't exist (headless-Chrome PDF rendering, for instance), recon shells out to a named external CLI ( `agent-browser` ) rather than linking the external dependency.
  3. **Protocol-rich**. HTTP(S), FTP(S), SFTP, TFTP, Gopher, SMTP(S), POP3(S), IMAP(S), SSH, SCP, Telnet, LDAP(S), WS(S), RTSP(S), Dict, NTP, Memcached, Redis, MQTT(S), IPFS/IPNS, Unix sockets, plus ping/traceroute.
  4. **Scriptable**. A Rhai script engine exposes every protocol probe, plus cryptography, compression, barcodes, JSON/YAML/XML handling, SQLite, file I/O, and multi-threaded concurrency. Scripts can stand in for shell pipelines and integration-test harnesses.
  5. **Diagnostic-friendly**. Verbose output, curl-style write-out, prettified JSON/XML, precise error messages, exit codes that match curl's.
-

# Installation

---

## From source

---

```
git clone https://github.com/thomas-starweb/recon
cd recon
cargo build --release
cp target/release/recon ~/.local/bin/      # or /usr/local/bin with sudo
```

You'll need:

- Rust 1.75 or newer ( `rustup install stable` ).
- A system linker; `cargo` handles the rest.
- For the optional `--html-to-pdf` / `--md-to-pdf` features: `agent-browser` on `$PATH` ( `brew install agent-browser` on macOS, or `npm install -g agent-browser` elsewhere ).

## First-run bootstrap

---

```
recon --init
```

creates `~/.recon/` with `script/`, `jars/`, `sni/`, and a commented `config.toml` skeleton. Existing files are never overwritten. See the [~/.recon/ layout](#) appendix.

---

## Quick start

---

```

# HTTP(S)
recon https://api.example.com/status
recon -X POST https://api.example.com/users -d '{"name":"a"}' -H 'Content-Type: application/json'
recon --json '{"q":"rust"}' https://api.example.com/search # auto-sets headers
recon -L https://short.url/foo # follow redirects
recon https://site.example.com --LHEAD # follow
+ show each hop

# Inspection
recon https://example.com --cert # inspect TLS cert
recon example.com --dns #
A/AAAA/MX/TXT/NS/CNAME/SOA
recon --ping 8.8.8.8 # TCP + ICMP ping
recon --traceroute 8.8.8.8
recon --whois example.com

# Email-protection
recon example.com --spf --dmarc --dkim selector1 --mta-sts --bimi --tls-rpt

# Barcodes
recon --encode qr -o out.png 'https://example.com'
recon --decode out.png # →
qr<TAB>https://example.com

# Conversions
recon --md-to-pdf README.md --toc --gfm -o README.pdf
recon --compare a.json b.json # GNU-diff compatible

# Scripting
recon --script my-probe.rhai https://api.example.com
recon --init #
bootstrap ~/.recon/

# Help surface
recon --help # clap-generated flag list
recon --flags # curl-style alphabetical index
recon --help http # deep-dive on HTTP topic
recon --help script # deep-dive on scripting

```

```
recon --examples # curated
examples, paged
recon --version #
protocols + features
```

---

# How recon is organized

---

recon has one binary with many modes. The mode is selected by flag:

| Mode group             | Selector flags                                                                                                                                                              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HTTP request (default) | no mode flag, or a URL positional argument                                                                                                                                  |
| Source-inspection      | <code>--hash</code> , <code>--compress</code> , <code>--decompress</code> , <code>--encode</code> , <code>--decode</code> , <code>--encrypt</code> , <code>--decrypt</code> |
| Email-protection       | <code>--spf</code> , <code>--dmarc</code> , <code>--dkim</code> , <code>--mta-sts</code> , <code>--bimi</code> , <code>--tls-rpt</code>                                     |
| Network tests          | <code>--ping</code> , <code>--traceroute</code> , <code>--netstatus</code> , <code>--whois</code>                                                                           |
| Certificate inspection | <code>--cert</code> , <code>--dns</code> , or positional URL with <code>--cert</code> / <code>--dns</code>                                                                  |
| SMTP send              | <code>--mail-from</code> and <code>--mail-to</code>                                                                                                                         |
| Serve                  | <code>--serve</code> , <code>--serve-tls</code>                                                                                                                             |
| JWT                    | <code>--jwt-view</code> , <code>--jwt-sign</code> , <code>--jwt-validate</code>                                                                                             |
| Archive                | <code>--archive</code> , <code>--extract</code>                                                                                                                             |
| Checkdigit             | <code>--checkdigit</code> , <code>--checkdigit-create</code> , <code>--checkdigit-list</code>                                                                               |
| Sample                 | <code>--sample</code> , <code>--sample-list</code>                                                                                                                          |
| Browser                | <code>--browser-screenshot</code>                                                                                                                                           |
| Compare                | <code>--compare</code>                                                                                                                                                      |
| Docs                   | <code>--md-to-html</code> , <code>--md-to-pdf</code> , <code>--html-to-pdf</code>                                                                                           |
| Script                 | <code>--script</code> (turns recon into a Rhai interpreter)                                                                                                                 |
| Meta                   | <code>--init</code> , <code>--editor-cleanup</code> , <code>--list-charsets</code> , <code>--iconv</code>                                                                   |

Only the modes in the first group do an HTTP request. Every other mode is stand-alone.

---

# Global conventions

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- **-h, --help** prints the clap-generated flag summary. **--help <topic>** routes to a deep-dive help topic (see `recon --help` footer for the list).
  - **-V, --version** prints the curl-compatible multi-line banner. **--version-short** prints just the version number.
  - **-v** increases verbosity. Repeat ( **-vv** , **-vvv** ) for more detail. At default, connection lines and protocol handshakes are suppressed.
  - **-s, --silent** suppresses progress + verbose output. **--show-error** re-enables error output when **-s** is set.
  - **-o <FILE>** writes the response body to a file instead of stdout. **-O** / **--remote-name** derives the filename from the URL's path. **-J** / **--remote-header-name** respects Content-Disposition.
  - **-f, --fail** exits non-zero on HTTP 4xx/5xx (no body printed). **--fail-with-body** prints the body but still exits non-zero.
  - **-k, --insecure** disables TLS cert verification. Use for self-signed tests; never in production scripts.
  - **-p, --pretty** pretty-prints JSON / XML / YAML bodies. Auto-detection by Content-Type; can be disabled with **--no-pretty**.
  - **-L, --location** follows HTTP redirects. **--max-redirs N** caps the chain (default 10). **--LHEAD** follows and prints each hop's headers.
  - **-A "..."** sets the User-Agent. Default is `recon/<version>`.
  - **-e <URL>** sets the Referer header.
  - **-H 'Name: Value'** adds a request header. Repeat for multiple.
  - **--max-time <SECS>** overall operation timeout. **--connect-timeout <SECS>** TCP connect timeout (default 30).
  - **-w '<FORMAT>'** writes a curl-compatible write-out string to stdout after the body. See [Write-out format](#).
  - **--json <DATA>** curl's **--json** compatibility: sends DATA as a JSON body and auto-sets Content-Type: `application/json` + `Accept: application/json`.
  - **-T <FILE>** uploads the file as the request body (default method PUT).
  - **Environment variables** — many flags honor the same env var curl would ( `HTTP_PROXY` , `HTTPS_PROXY` , `NO_PROXY` , `ALL_PROXY` , `SSL_CERT_FILE` , `CURL_CA_BUNDLE` ). `recon`-specific ones are prefixed `RECON_`. See [Environment variables](#).
-

# Part II — CLI reference

## HTTP / HTTPS requests

The default mode. A positional URL (or `--url <URL>`) triggers an HTTP request through the request + rustls stack.

**0.62.0 curl-parity additions** — see the curl easy-wins section of `recon --examples` and the relevant flags table rows below. Quick summary: `-r/--range`, `-z/--time-cond`, `--etag-compare`, `--etag-save`, `--timestamping`, `--max-filesize`, `--url-query`, `--disallow-username-in-url`, `--remove-on-error`, `--no-clobber`, `--create-file-mode`, `-N/--no-buffer`, `-D/--dump-header`, `--stderr`, `--no-progress-meter`, `--tcp-nodelay`, `--no-keepalive`, `--keepalive-time`, `--capath`, `--ca-native`, `--tls-max`, `--connect-to`, `--oauth2-bearer`, `--xattr`, `--spider`.

### Core flags

| Flag                                          | Description                                                                                                                 |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <code>-X, --request &lt;METHOD&gt;</code>     | GET, POST, PUT, PATCH, DELETE, HEAD. Default GET (PUT when <code>-T</code> set, POST when <code>-d</code> set).             |
| <code>-H, --header &lt;NAME: VALUE&gt;</code> | Add a request header. Repeatable.                                                                                           |
| <code>-d, --data &lt;BODY   @FILE&gt;</code>  | Request body. <code>@file</code> reads from disk, <code>@-</code> reads stdin. Promotes GET → POST unless <code>-G</code> . |
| <code>--json &lt;DATA&gt;</code>              | Curl-compatible: sends DATA as JSON, sets <code>Content-Type: application/json</code> + <code>Accept</code> .               |
| <code>--data-raw &lt;DATA&gt;</code>          | Like <code>-d</code> but <code>@file</code> is NOT processed (sends literal string).                                        |
| <code>--data-binary &lt;DATA&gt;</code>       | Like <code>-d</code> but CR/LF are not stripped from <code>@file</code> content.                                            |
| <code>--data-urlencode &lt;DATA&gt;</code>    | URL-encode DATA. Repeatable; values joined with <code>&amp;</code> .                                                        |
| <code>-G, --get</code>                        | Send <code>-d</code> data as query parameters on GET (body empty).                                                          |
| <code>-T, --upload-file &lt;PATH&gt;</code>   | Upload file as request body. Default PUT.                                                                                   |
| <code>-L, --location</code>                   | Follow redirects (3xx).                                                                                                     |

| Flag                                          | Description                                                 |
|-----------------------------------------------|-------------------------------------------------------------|
| <code>--max-redirs &lt;N&gt;</code>           | Cap redirect chain (default 10).                            |
| <code>--LHEAD</code>                          | Follow redirects and print each hop's headers.              |
| <code>-e, --referer &lt;URL&gt;</code>        | Set Referer header. <code>--referrer</code> alias accepted. |
| <code>-A, --user-agent &lt;STR&gt;</code>     | User-Agent. Default <code>recon/&lt;version&gt;</code> .    |
| <code>--compressed</code>                     | Request + auto-decompress gzip/deflate/brotli/zstd.         |
| <code>--max-time &lt;SECS&gt;</code>          | Total operation timeout (seconds, fractional OK).           |
| <code>--connect-timeout &lt;SECS&gt;</code>   | TCP connect timeout (default 30).                           |
| <code>-f, --fail</code>                       | Exit non-zero on HTTP 4xx/5xx, suppress body.               |
| <code>--fail-with-body</code>                 | Exit non-zero on 4xx/5xx but still print body.              |
| <code>-4, --ipv4 / -6, --ipv6</code>          | Force IPv4 / IPv6 resolution.                               |
| <code>--resolve &lt;HOST:PORT:ADDR&gt;</code> | Override DNS for a specific host:port → addr. Repeatable.   |

## Examples

```

# Simple requests
recon https://httpbin.org/get
recon -X POST https://api.example.com/v1/users -d '{"name":"a"}' -H
'Content-Type: application/json'
recon --json '{"query":"rust"}' https://api.example.com/search

# Body from files / stdin
recon https://upload.example.com -d @payload.json -H 'Content-Type:
application/json'
cat body.json | recon -X POST https://api.example.com -d @-

# Upload
recon -T ./report.pdf https://upload.example.com/reports/          # PUT by
default
recon -T ./report.pdf https://upload.example.com/ -X POST          # override
method

# Follow redirects
recon -L https://bitly.example.com/abc
recon --LHEAD https://example.com/                                # show each
hop's headers

# Fail fast
recon -f https://api.example.com/users/42                          # exit 22
on 4xx/5xx, no body
recon --fail-with-body https://api.example.com/users/42           # same but
still print body

# Rate and timeout
recon --max-time 10 https://slow.example.com/
recon --connect-timeout 3 --max-time 8 https://flaky.example.com/
recon --limit-rate 500K https://download.example.com/big.bin -o big.bin
recon --speed-limit 10000 --speed-time 30
https://stalling.example.com/big.bin -0

```

## Wget-style batch fetching

Long-form wget-compatible flags for batch URL handling. Short forms (`-A`, `-R`, `-w`, `-t`, `-r`, `-l`, `-m`, `-p`, `-k`, `-D`, `-H`, `-np`) are intentionally not provided — recon reserves single-letter flags for curl compatibility. The recursive/mirror cluster (`-r`, `-l`, `-m`, `-p`, `-k`, etc.) is deferred and tracked in `OUT-OF-SCOPE.md`.

| Flag                                                   | Description                                                                                                                                       |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--input-file</code><br><code>&lt;FILE&gt;</code> | Batch-fetch URLs listed in FILE (one per line, <code>#</code> comments, blank lines ignored, <code>-</code> reads from stdin). Each URL processed |

| Flag                                      | Description                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | independently.                                                                                                                                                                                                                             |
| <code>--wait &lt;SECS&gt;</code>          | (0.67.0) Fixed-seconds delay between URLs in batch mode. Skipped before the first URL. Overrides <code>--rate</code> when both are set.                                                                                                    |
| <code>--tries &lt;N&gt;</code>            | (0.67.0) Total attempts per URL (wget semantics: <code>tries = retries + 1</code> ). Overrides <code>--retry</code> . <code>--tries 1</code> disables retries; <code>--tries 0</code> is rejected.                                         |
| <code>--accept &lt;LIST&gt;</code>        | (0.67.0) Comma-separated filename-suffix accept list (case-insensitive). e.g. <code>--accept jpg,png</code> keeps only URLs whose final path segment ends in <code>.jpg</code> or <code>.png</code> . URLs with empty final segments fail. |
| <code>--reject &lt;LIST&gt;</code>        | (0.67.0) Comma-separated filename-suffix reject list. Combines with <code>--accept</code> (URL must pass both). URLs with empty final segments pass.                                                                                       |
| <code>--continue</code>                   | Resume an interrupted download (wget-style auto-detect from local file size). Equivalent to <code>--continue-at -</code> .                                                                                                                 |
| <code>--continue-at &lt;OFFSET&gt;</code> | Resume from BYTE offset (curl-compatible). <code>-</code> auto-detects.                                                                                                                                                                    |
| <code>--spider</code>                     | HEAD-only check; print <code>&lt;status&gt; &lt;url&gt;</code> per URL and exit non-zero on any 4xx/5xx. Pairs with <code>--input-file</code> .                                                                                            |
| <code>--timestamping</code>               | Skip download when local file's mtime $\geq$ server's Last-Modified. Sets If-Modified-Since.                                                                                                                                               |
| <code>--rate &lt;N/s N/m N/h&gt;</code>   | Request rate cap. Engages with <code>--input-file</code> : at most N requests per second/minute/hour. Overridden by <code>--wait</code> .                                                                                                  |

## Examples

```
# Polite batch fetch with a 2-second gap between URLs
recon --input-file urls.txt --wait 2

# Filter to image URLs, drop thumbnails
recon --input-file urls.txt --accept jpg,png --reject thumb

# Wget-style retries (5 total attempts)
recon https://api.example.com/ --tries 5

# Spider check filtered by extension
recon --input-file urls.txt --spider --accept html,htm

# Resume an interrupted download
recon https://example.com/big.iso -o big.iso --continue
```

## Output control

| Flag                                        | Description                                                                             |
|---------------------------------------------|-----------------------------------------------------------------------------------------|
| <code>-o, --output &lt;FILE&gt;</code>      | Write body to FILE (otherwise stdout).                                                  |
| <code>-O, --remote-name</code>              | Filename = URL's last path segment (percent-decoded).                                   |
| <code>-J, --remote-header-name</code>       | Use Content-Disposition filename when saving.                                           |
| <code>--create-dirs</code>                  | Create parent dirs for <code>-o</code> path.                                            |
| <code>-i, --include</code>                  | Print response headers before body.                                                     |
| <code>-I, --head</code>                     | Print headers only; no body. Implies <code>-X HEAD</code> .                             |
| <code>-s, --silent</code>                   | Suppress progress + verbose output.                                                     |
| <code>--show-error</code>                   | Re-enable error output even when <code>-s</code> is set.                                |
| <code>-v, --verbose</code>                  | Verbose. Repeatable: <code>-v</code> , <code>-vv</code> , <code>-vvv</code> .           |
| <code>-p, --pretty</code>                   | Pretty-print JSON / XML / YAML bodies.                                                  |
| <code>--no-pretty</code>                    | Disable auto-pretty (even when content-type suggests it).                               |
| <code>-w, --write-out &lt;FORMAT&gt;</code> | Print a curl-compatible summary after the body. See <a href="#">Write-out format</a> .  |
| <code>--editor</code>                       | Open response body in <code>\$EDITOR</code> after the request.                          |
| <code>--json</code> (output-side inferred)  | Combined with <code>-p</code> , forces JSON pretty-printing regardless of content-type. |

## Examples

```

# Save to file
recon https://example.com/favicon.ico -o          # →
favicon.ico
recon https://example.com/api/users.json -o users.json
recon https://x.example.com/deep/path/file.txt --create-dirs -o
downloads/x/file.txt

# Inspect headers
recon -I https://example.com/                      # HEAD
recon -i https://api.example.com/v1/status         # headers +
body
recon -i -I https://example.com/                   # forced
HEAD with body suppressed

# Pretty-print
recon https://api.example.com/large.json -p
recon https://api.example.com/feed.xml -p          # XML also
supported
curl -s https://api.example.com/blob.yaml | recon -p - # stdin
source

# Verbose progression
recon -v https://api.example.com/                  # connection + TLS + headers
recon -vv https://api.example.com/                 # plus intermediate request info
recon -vvv https://api.example.com/                 # plus raw handshake + wire dump

# Writeout
recon https://api.example.com/ -w '\n%{http_code} in %{time_total}s (%
{size_download} bytes)\n'

```

## Authentication & TLS

| Flag                                      | Description                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-u, --user &lt;USER:PASS&gt;</code> | HTTP Basic credentials.                                                                                                                           |
| <code>-k, --insecure</code>               | Skip TLS cert verification.                                                                                                                       |
| <code>--tlsv1.2 / --tlsv1.3</code>        | Force minimum TLS version.                                                                                                                        |
| <code>--cacert &lt;PATH&gt;</code>        | Trust an extra PEM root (on top of system roots).                                                                                                 |
| <code>--interface &lt;IP NAME&gt;</code>  | Bind outgoing socket to a specific local IP <b>or</b> interface name (eth0, en0 on Linux/macOS via getifaddrs; Windows accepts IP literals only). |

| Flag                                                     | Description                                         |
|----------------------------------------------------------|-----------------------------------------------------|
| <code>--limit-rate</code><br><code>&lt;RATE&gt;</code>   | Throttle download. Suffixes: K, M, G.               |
| <code>--speed-limit</code><br><code>&lt;BYTES&gt;</code> | Minimum bytes-per-second (fails if rate drops).     |
| <code>--speed-time</code><br><code>&lt;SECS&gt;</code>   | Window for <code>--speed-limit</code> (default 30). |

## Examples

```
recon -u alice:s3cr3t https://private.example.com/
recon -k https://self-signed.example.com/
recon --tlsv1.3 https://example.com/           # refuse downgrade to 1.2
recon --cacert /etc/corp-root.pem https://internal.corp/
recon --interface 10.0.0.5 https://example.com/ # use a specific source IP
```

## Client certificates (mTLS)

Shipped in 0.54.0. Present a client certificate during the TLS handshake for mutual TLS.

| Flag                                                        | Description                                                                                                    |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>-E, --client-cert</code><br><code>&lt;PATH&gt;</code> | PEM-encoded client cert. May include the key inline.                                                           |
| <code>--client-key</code><br><code>&lt;PATH&gt;</code>      | PEM key (when cert is cert-only).                                                                              |
| <code>--cert-type</code><br><code>&lt;PEM DER&gt;</code>    | Cert format. DER errors with a conversion recipe.                                                              |
| <code>--key-type</code><br><code>&lt;PEM DER ENG&gt;</code> | Key format. ENG refused (rustls has no engine concept).                                                        |
| <code>--pass &lt;PASS&gt;</code>                            | Placeholder for encrypted PKCS#8 passphrase. Encrypted keys refused with an <code>openssl pkcs8</code> recipe. |

## Examples

```
# Combined cert+key PEM
recon -E ~/keys/bundle.pem https://mtls.example.com/

# Split cert and key
recon -E client.crt --client-key client.key https://mtls.example.com/

# Decrypt encrypted PKCS#8 externally first
openssl pkcs8 -in client.key.enc -out client.key
recon -E client.crt --client-key client.key https://mtls.example.com/
```

See also: `recon --help client-cert`.

## Proxy routing

0.50.0 shipped the full proxy suite. Schemes auto-detected from the URL: `http://`, `https://`, `socks5://`, `socks5h://`.

| Flag                                            | Description                                                                                                  |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>-x, --proxy &lt;URL&gt;</code>            | Proxy URL. Falls back to <code>\$HTTPS_PROXY</code> / <code>\$HTTP_PROXY</code> / <code>\$ALL_PROXY</code> . |
| <code>-U, --proxy-user &lt;USER:PASS&gt;</code> | Basic auth for the proxy.                                                                                    |
| <code>--noproxy &lt;LIST&gt;</code>             | Comma-separated bypass list (matches <code>\$NO_PROXY</code> semantics; <code>*</code> means bypass all).    |
| <code>--proxy-insecure</code>                   | Skip cert verification on the TLS-to-proxy connection.                                                       |
| <code>--proxy-cacert &lt;PATH&gt;</code>        | Extra CA for the proxy connection.                                                                           |

## Examples

```

recon -x http://proxy.corp:3128 https://api.example.com/
recon -x socks5h://127.0.0.1:9050 https://example.onion/
recon -x https://proxy.example.com:8443 -U alice:s3cr3t
https://example.com/
HTTPS_PROXY=http://proxy.corp:3128 NO_PROXY='.internal' recon
https://api.example.com/

# Script equivalent
recon --script - <<'EOF'
http("https://api.example.com/", #{
  proxy: "socks5h://127.0.0.1:9050",
  proxy_user: "alice:s3cr3t",
  noproxy: ".internal",
});
EOF

```

## Unix-domain sockets

0.51.0. Route the HTTP request over a Unix socket instead of TCP. Host header and path are preserved; only the transport changes.

| Flag                                    | Description  |
|-----------------------------------------|--------------|
| <code>--unix-socket &lt;PATH&gt;</code> | Socket path. |

## Examples

```

# Docker API
recon --unix-socket /var/run/docker.sock http://localhost/_ping
recon --unix-socket /var/run/docker.sock -p http://localhost/v1.40/version
recon --unix-socket /var/run/docker.sock
http://localhost/v1.40/containers/json

# systemd-activated service
recon --unix-socket /run/app.sock http://localhost/api/v1/status

```

## HSTS persistent cache

0.52.0. A curl-compatible HSTS cache that upgrades `http://` to `https://` before sending (when the host has a non-expired cache entry) and updates itself from `Strict-Transport-Security` response headers.

| Flag                             | Description                                |
|----------------------------------|--------------------------------------------|
| <code>--hsts &lt;FILE&gt;</code> | Load + update the HSTS cache at this path. |

## Examples

```
# Populate from an https:// response
recon --hsts ~/.recon/hsts.txt https://www.cloudflare.com/

# Future http:// requests to the same host are auto-upgraded
recon --hsts ~/.recon/hsts.txt http://www.cloudflare.com/
# * HSTS: upgrading http:// to https:// for www.cloudflare.com

# File format matches curl's --hsts:
cat ~/.recon/hsts.txt
# # HSTS cache (recon 0.58.1)
# # host expires_unix (leading . = includeSubDomains)
# .www.cloudflare.com 1808493843
```

## IPFS / IPNS

0.49.0. `ipfs://<cid>` and `ipns://<name>` URLs are rewritten to an HTTP gateway URL before the request fires.

| Flag                                    | Description                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>--ipfs-gateway &lt;URL&gt;</code> | Gateway base URL. Default <code>https://ipfs.io</code> . Also read from <code>\$RECON_IPFS_GATEWAY</code> . |

## Examples

```
recon ipfs://QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG
recon --ipfs-gateway http://127.0.0.1:8080 ipfs://bafy... # local Kubo
node
recon ipns://example.eth # via the
default gateway
```

## Cookies

| Flag                                      | Description                                    |
|-------------------------------------------|------------------------------------------------|
| <code>-b, --cookiejar &lt;PATH&gt;</code> | SQLite cookie jar at PATH (created on demand). |
| <code>--cookies</code>                    | Read-only listing of the current cookie jar.   |

| Flag                                       | Description              |
|--------------------------------------------|--------------------------|
| <code>--cookie-set &lt;NAME=VAL&gt;</code> | Set a cookie in the jar. |
| <code>--cookie-delete &lt;NAME&gt;</code>  | Remove by name.          |

Cookies persist across invocations when you pass the same `--cookiejar` path. The `.db` format is SQLite; inspect with `sqlite3` or any sqlite viewer.

## Examples

```
# Login, save cookies, browse with them
recon https://example.com/login -d 'user=alice&pass=secret' --cookiejar ~/jar.db
recon https://example.com/dashboard --cookiejar ~/jar.db -p

# Inspect
recon --cookiejar ~/jar.db --cookies

# Inject manually
recon --cookiejar ~/jar.db --cookie-set 'session=abc123;
Domain=.example.com; Path=/'
```

## DNS

| Flag                                      | Description                                                                                                                                  |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--dns</code>                        | Resolve A, AAAA, MX, TXT, NS, CNAME, SOA for the URL's host.                                                                                 |
| <code>--dns-type &lt;LIST&gt;</code>      | Comma-separated subset:<br>A, AAAA, MX, TXT, NS, CNAME, SOA, PTR, CAA, DS, DNSKEY, HTTPS, SRV, SVCB.                                         |
| <code>--dns-servers &lt;LIST&gt;</code>   | Comma-separated resolver IPs.                                                                                                                |
| <code>--dns-ipv4-addr &lt;IP&gt;</code>   | Bind DNS queries to a specific local IPv4.                                                                                                   |
| <code>--dns-ipv6-addr &lt;IP&gt;</code>   | Same for IPv6.                                                                                                                               |
| <code>--dns-interface &lt;NAME&gt;</code> | Accepted at the CLI; not yet plumbed (see OUT-OF-SCOPE.md). Use <code>--dns-ipv4-addr</code> / <code>--dns-ipv6-addr</code> as a workaround. |

## Examples

```
recon example.com --dns
recon example.com --dns-type A,AAAA,MX
recon example.com --dns --dns-servers 1.1.1.1,8.8.8.8
recon example.com --dns --dns-servers 127.0.0.1:5353      # local
resolver on custom port
```

## TLS certificate inspection

Standalone mode selected by `--cert` (without a `--cert <PATH>` value — it's the bool form of the flag).

| Flag                                                | Description                                                                    |
|-----------------------------------------------------|--------------------------------------------------------------------------------|
| <code>--cert</code>                                 | Inspect the server cert: subject, issuer, SANs, NotBefore/After, fingerprints. |
| <code>--cert-chain</code>                           | Walk the full intermediate chain.                                              |
| <code>--sni</code><br><code>&lt;HOSTNAME&gt;</code> | Override the SNI value sent in the handshake.                                  |

## Examples

```
recon https://example.com --cert
recon https://example.com --cert --cert-chain
recon https://example.com --cert --sni alt.example.com      # request
cert for alt via SNI
```

## Ping, traceroute, netstatus

Stand-alone modes.

| Flag                                                   | Description                                                           |
|--------------------------------------------------------|-----------------------------------------------------------------------|
| <code>--ping &lt;HOST&gt;</code>                       | TCP ping by default, ICMP with <code>--icmp</code> .                  |
| <code>--icmp</code>                                    | Force ICMP (needs raw sockets; root on Linux, CAP_NET_RAW elsewhere). |
| <code>--ping-count &lt;N&gt;</code>                    | Packets (default 4).                                                  |
| <code>--traceroute</code><br><code>&lt;HOST&gt;</code> | Same as <code>--ping HOST --traceroute</code> .                       |

| Flag                              | Description                                                                     |
|-----------------------------------|---------------------------------------------------------------------------------|
| <code>--max-hops &lt;N&gt;</code> | Hop limit (default 30).                                                         |
| <code>--netstatus</code>          | Run the probe bundle defined in <code>~/.recon/config.toml</code> [netstatus] . |

## Examples

```
recon --ping 8.8.8.8
recon --ping example.com --ping-count 10
recon --ping example.com:443 --icmp          # ICMP with explicit port hint
recon --traceroute 8.8.8.8 --max-hops 20
recon --netstatus                          # connectivity sweep
```

## Email protection

A single URL can be probed for its whole email-auth stack in one invocation.

| Flag                                 | Description                                                      |
|--------------------------------------|------------------------------------------------------------------|
| <code>--spf</code>                   | SPF record validation (recursive include/redirect, lookup limit) |
| <code>--dmarc</code>                 | DMARC policy + record                                            |
| <code>--dkim &lt;SELECTOR&gt;</code> | DKIM for the given selector. Repeatable.                         |
| <code>--mta-sts</code>               | MTA-STS policy                                                   |
| <code>--bimi [SELECTOR]</code>       | BIMI record + optional VMC / SVG fetch                           |
| <code>--tls-rpt</code>               | SMTP TLS-RPT record                                              |

## Examples

```
# Full sweep
recon example.com --spf --dmarc --dkim selector1 --mta-sts --bimi --tls-rpt

# Single checks
recon example.com --spf
recon example.com --dkim selector1 --dkim selector2
recon example.com --bimi default          # explicit selector
```

## SMTP

Stand-alone SMTP client. Send mail or just probe capabilities.

| Flag                                       | Description                                                |
|--------------------------------------------|------------------------------------------------------------|
| <code>--mail-from &lt;ADDR&gt;</code>      | Envelope sender. Required for send.                        |
| <code>--mail-to &lt;ADDR&gt;</code>        | Envelope recipient. Repeatable.                            |
| <code>--mail-subject &lt;STR&gt;</code>    | Subject header. Default "recon SMTP test".                 |
| <code>--mail-body &lt;STR&gt;</code>       | Body. Accepts <code>@file</code> , <code>@-</code> .       |
| <code>--mail-header &lt;H: V&gt;</code>    | Extra header. Repeatable.                                  |
| <code>--smtp-auth &lt;USER:PASS&gt;</code> | AUTH PLAIN → LOGIN chain.                                  |
| <code>--smtp-helo &lt;NAME&gt;</code>      | HELO/EHLO name. Default <code>recon.local</code> .         |
| <code>--no-starttls</code>                 | Don't negotiate STARTTLS.                                  |
| <code>--dkim-key &lt;PATH&gt;</code>       | DKIM-sign outbound with this PEM key.                      |
| <code>--dkim-selector &lt;SEL&gt;</code>   | DKIM selector.                                             |
| <code>--dkim-domain &lt;DOM&gt;</code>     | DKIM domain (defaults to <code>--mail-from</code> domain). |

## Examples

```
# Probe only
recon smtp://mail.example.com:25

# Send
recon smtp://smtp.example.com:587 --mail-from me@example.com --mail-to
you@example.com \
    --mail-subject 'Hi' --mail-body 'Test' --smtp-auth me:s3cr3t

# With DKIM signing
recon smtp://smtp.example.com:587 --mail-from me@example.com --mail-to
you@example.com \
    --dkim-key ~/.dkim/default.pem --dkim-selector default
```

## Mail retrieval (POP3, IMAP)

| Flag                     | Description                             |
|--------------------------|-----------------------------------------|
| <code>--stls</code>      | POP3: upgrade via STLS after CAPA.      |
| <code>--imap-peek</code> | IMAP: use BODY.PEEK (don't flip \Seen). |

URLs: `pop3://`, `pop3s://`, `imap://`, `imaps://`.

## Examples

```

recon pop3s://alice:s3cr3t@pop.example.com/
recon pop3://alice:s3cr3t@pop.example.com/ --stls
recon imaps://alice:s3cr3t@imap.example.com/INBOX
recon imaps://alice:s3cr3t@imap.example.com/INBOX/3          # fetch message
3
recon imaps://alice:s3cr3t@imap.example.com/INBOX --imap-peek

```

## File transfer

0.47.0 shipped FTP/FTPS, SFTP, TFTP, Gopher.

| Protocol      | URL scheme               | Notes                                                                      |
|---------------|--------------------------|----------------------------------------------------------------------------|
| FTP /<br>FTPS | ftp://, ftps://          | Anonymous by default; userinfo in URL or <code>-u</code> .                 |
| SFTP          | sftp://                  | SSH-backed. <code>--privkey</code> for a specific key.                     |
| TFTP          | tftp://                  | RFC 1350. <code>--tftp-blksize</code> for RFC 2348 block-size negotiation. |
| Gopher        | gopher://,<br>gophers:// | Selector fetch.                                                            |

## Examples

```

# FTP - listing vs fetch
recon ftp://ftp.example.com/                                     #
directory listing
recon ftp://ftp.example.com/incoming/file.bin -o local.bin      #
retrieve
recon ftp://alice:s3cr3t@ftp.example.com/                       #
authenticated

# SFTP
recon sftp://me@example.com/home/me/data.csv -o data.csv
recon sftp://me@example.com:2222/srv/ --privkey ~/.ssh/ci_rsa

# TFTP
recon tftp://tftp.example.com/pxelinux.cfg/default
recon tftp://tftp.example.com/boot.img -o boot.img --tftp-blksize 8192

# Gopher
recon gopher://gopher.floodgap.com/
recon gopher://gopher.floodgap.com/0/gopher/proxy

```

# Other protocols

---

## SSH, SCP

```
recon ssh://me@example.com -- uname -a          # SSH exec; prints stdout
recon scp://me@example.com/etc/motd -o motd      # SCP fetch
```

## Telnet

```
recon telnet://router.example.com:23
```

## LDAP

```
recon ldap://ldap.example.com -H 'Bind: cn=admin,dc=example,dc=com'
recon ldaps://ldap.example.com          # implicit-TLS
```

## WebSocket

```
recon wss://echo.websocket.events      # handshake + Ping/Pong
```

## RTSP

```
recon rtsp://camera.example.com/stream1  # OPTIONS
```

## Dict (RFC 2229)

```
recon dict://dict.org/d:recon
```

## NTP, Memcached, Redis, MQTT

```
recon ntp://pool.ntp.org                # clock offset + rtt
recon memcache://cache.example.com/?stats # text-protocol stats
recon redis://cache.example.com/         # PING
recon redis://cache.example.com/ -X INFO  # arbitrary RESP command
recon mqtt://broker.example.com --mqtt-topic status --mqtt-message 'hello'
recon mqtt://broker.example.com --mqtt-user alice --mqtt-pass s3cr3t ...
```

## Source comparison

0.53.0 shipped `--compare` . Diff two sources (URL / path / `-` stdin). HTTP(S) sources flow through the full request pipeline so every request flag (`-H`, `-u`, `-L`, `-k`, cookies, proxy, HSTS) applies.

| Flag                                       | Description                                                               |
|--------------------------------------------|---------------------------------------------------------------------------|
| <code>--compare &lt;A&gt; &lt;B&gt;</code> | Two sources.                                                              |
| <code>--compare-format &lt;FMT&gt;</code>  | <code>unified</code> (default), <code>summary</code> , <code>sxs</code> . |
| <code>--compare-context &lt;N&gt;</code>   | Unified context lines (default 3).                                        |

Exit codes follow GNU `diff` : 0 identical, 1 differ, 2+ source-load error.

## Examples

```
recon --compare one.json two.json
recon --compare https://api.example.com/v1/status ./baseline.json
curl -s https://a/ | recon --compare - ./baseline.json
recon --compare a.txt b.txt --compare-format summary
recon --compare a.txt b.txt --compare-format sxs
recon --compare a.json b.json --compare-context 5
```

## Document conversions

0.58.0 shipped markdown / HTML / PDF. See also `--help docs` .

| Flag                                   | Description                                                        |
|----------------------------------------|--------------------------------------------------------------------|
| <code>--md-to-html &lt;SRC&gt;</code>  | Markdown → HTML via comrak. Output <code>-o PATH</code> or stdout. |
| <code>--md-to-pdf &lt;SRC&gt;</code>   | Markdown → PDF (via agent-browser). <code>-o PATH</code> required. |
| <code>--html-to-pdf &lt;SRC&gt;</code> | HTML → PDF (via agent-browser).                                    |
| <code>--toc</code>                     | Inject a linkable TOC.                                             |
| <code>--toc-depth &lt;N&gt;</code>     | Include headings up to H <code>N</code> (default 3).               |
| <code>--toc-title &lt;STR&gt;</code>   | TOC heading (default "Contents").                                  |

| Flag                                 | Description                                                                                                                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--doc-title &lt;STR&gt;</code> | <code>&lt;title&gt;</code> + PDF metadata title.                                                                                                                                                                  |
| <code>--doc-css &lt;PATH&gt;</code>  | Inline a custom stylesheet.                                                                                                                                                                                       |
| <code>--no-default-css</code>        | Skip bundled default CSS.                                                                                                                                                                                         |
| <code>--gfm</code>                   | Enable GitHub-flavored extensions (tables, task lists, strikethrough, autolinks, footnotes, tagfilter).                                                                                                           |
| <code>--unsafe-html</code>           | Allow raw HTML passthrough (comrak's <code>unsafe_</code> flag). Needed for cover pages and explicit <code>&lt;div class="page-break"&gt;</code> markers. Off by default; assume the markdown is trusted when on. |
| <code>--page-break-on-h1</code>      | Start a new PDF page before every top-level <code>#</code> heading except the first. No visible effect in HTML output (Chrome's printToPDF honours the <code>break-before: page</code> rule).                     |

## Examples

```
recon --md-to-html README.md --toc --gfm -o README.html
recon --md-to-pdf CHANGELOG.md --toc --gfm --doc-title 'recon release notes' -o changelog.pdf
recon --html-to-pdf report.html -o report.pdf
curl -s https://example.com/doc.md | recon --md-to-html - --toc -o doc.html
recon --md-to-pdf notes.md --toc --doc-css print.css -o notes.pdf
recon --md-to-pdf notes.md --no-default-css --doc-css corp.css -o notes.pdf

# Book-style: cover page + chapter breaks
recon --md-to-pdf book.md --toc --gfm --unsafe-html --page-break-on-h1 \
    --doc-title 'My Book' -o book.pdf
```

## Cover pages

With `--unsafe-html`, raw HTML passes through the markdown parser verbatim. The bundled CSS styles `<div class="cover">` as a full-page centered block with an automatic page break after:

```

<div class="cover">

# My Document

<div class="subtitle">An illustrated guide</div>

<hr>

<div class="version">Version 1.0</div>
<div class="date">2026-04-24</div>
<div class="author">Alice Example</div>

</div>

# First chapter

Body text...

```

The cover block uses a `min-height: 90vh` flex container with centered content. `.subtitle`, `.version`, `.date`, `.author`, and `.meta` child classes pick up smaller, muted styling. A horizontal rule becomes a narrow divider.

## Page breaks

Two routes:

1. **`--page-break-on-h1`** — every top-level `#` heading starts a new PDF page (except the first, which opens the document). No markdown changes required.
2. **Explicit markers (needs `--unsafe-html`)** — embed one of:

```

<div class="page-break"></div>
<hr class="page-break">

```

anywhere in the markdown. The CSS `break-after: page` kicks in.

Chrome's `printToPDF` is the renderer for both `--md-to-pdf` and `--html-to-pdf`, so any CSS that speaks `break-before`, `break-after`, or `page-break-*` works. `@page` rules for size and margins (defined in the bundled default CSS) are honored too — override via `--doc-css` or inline `<style>` with `--unsafe-html`.

## Barcode encoding & decoding

| Flag                                                     | Description                                                                                                             |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <code>--encode</code><br><code>&lt;FORMAT&gt;</code>     | qr, datamatrix, code128, code39, ean13, upca, aztec, pdf417.                                                            |
| <code>--encode-format</code><br><code>&lt;FMT&gt;</code> | ascii (default), svg, png. Inferred from <code>-o</code> extension.                                                     |
| <code>--from-file</code><br><code>&lt;PATH&gt;</code>    | Encode input from file (mutually exclusive with positional).                                                            |
| <code>--encode-list</code>                               | List all supported formats.                                                                                             |
| <code>--qr-level</code><br><code>&lt;L M Q H&gt;</code>  | QR error-correction level (default M).                                                                                  |
| <code>--hrt / --no-hrt</code>                            | Human-readable text under 1D barcodes. Default on for EAN/UPC, off for Code128/39. SVG + ASCII only (PNG HRT deferred). |
| <code>--decode</code><br><code>&lt;IMAGE&gt;</code>      | Scan a PNG/JPEG/WebP for any supported format. <code>-</code> for stdin.                                                |
| <code>--decode-hints</code><br><code>&lt;LIST&gt;</code> | Comma-separated format restriction.                                                                                     |
| <code>--decode-all</code><br><code>&lt;IMAGE&gt;</code>  | Scan for every barcode in the image; one line per detection.                                                            |

## Examples

```
# Encode
recon --encode qr -o qr.png 'https://example.com'
recon --encode qr --qr-level H -o durable.png 'sticker text'
recon --encode pdf417 -o id.png 'stacked linear code'
recon --encode aztec -o ticket.png 'transit ticket'
recon --encode ean13 --encode-format svg '4006381333931' > product.svg
recon --encode qr --from-file msg.txt -o qr-of-file.png

# Decode
recon --decode ticket.png                # → aztec<TAB>transit
ticket
cat code.png | recon --decode -
recon --decode mystery.png --decode-hints qr,datamatrix
recon --decode bottle.jpg --decode-hints ean13
recon --decode-all sheet.png            # every code in the
image
```

## HRT (human-readable text under 1D barcodes)

EAN-13 and UPC-A get HRT by default; Code128 and Code39 don't unless you pass `--hrt` . Implemented for ASCII and SVG output. PNG output ignores the flag — PNG HRT is deferred pending bundled-font work.

```
recon --encode ean13 --encode-format svg '4006381333931' -o retail.svg
recon --encode code128 --hrt --encode-format svg 'SHIP-4711' -o box.svg
recon --encode ean13 --no-hrt '4006381333931' -o bare.svg
```

## Hashing

| Flag                                             | Description                                                            |
|--------------------------------------------------|------------------------------------------------------------------------|
| <code>--hash</code><br><code>&lt;ALGO&gt;</code> | md5, sha1, sha224, sha256, sha384, sha512, sha3_256, sha3_512, blake3. |
| <code>--hash-list</code>                         | List supported algorithms.                                             |

## Examples

```
recon --hash sha256 /etc/hosts
recon --hash sha256 https://example.com/file.iso           # hash a URL
cat payload.bin | recon --hash blake3 -
echo -n "hello" | recon --hash md5 -
```

## Compression & archiving

| Flag                                                       | Description                                                                   |
|------------------------------------------------------------|-------------------------------------------------------------------------------|
| <code>--compress &lt;ALGO&gt;</code>                       | gzip, deflate, brotli, zstd, bzip2, lz4, xz, snap.                            |
| <code>--decompress &lt;ALGO&gt;</code>                     | Same set; auto-detect via magic bytes if ALGO is <code>auto</code> .          |
| <code>--compress-list</code>                               | List supported algorithms.                                                    |
| <code>--compression-level</code><br><code>&lt;N&gt;</code> | 1..22 (depends on algo).                                                      |
| <code>--archive &lt;DEST&gt;</code>                        | Create zip/tar/tar.gz/tar.xz/tar.bz2. Positional args after DEST are sources. |
| <code>--extract &lt;SRC&gt;</code>                         | Extract an archive. <code>-o DIR</code> to choose destination.                |

## Examples

```

recon --compress gzip /etc/hosts -o hosts.gz
recon --decompress auto hosts.gz -o hosts.plain
recon --compress zstd --compression-level 19 big.bin -o big.zst

recon --archive backup.tar.gz ~/Documents ~/src
recon --extract backup.tar.gz -o restore/
recon --extract release.zip

```

## Encryption

age-based + PGP shell-out.

| Flag                                           | Description                                                                                             |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <code>--encrypt</code>                         | Encrypt stdin / file to age format. Requires <code>--recipient</code> or <code>--pgp-recipient</code> . |
| <code>--decrypt</code>                         | Decrypt age / PGP. Needs <code>--identity</code> or a configured GPG keyring.                           |
| <code>--encrypt-keygen</code>                  | Generate a new age keypair.                                                                             |
| <code>--recipient &lt;KEY&gt;</code>           | Age recipient (X25519 or ssh-ed25519). Repeatable.                                                      |
| <code>--identity &lt;PATH&gt;</code>           | Age identity file for decryption.                                                                       |
| <code>--passphrase</code>                      | Use passphrase-based (scrypt) encryption. Mutually exclusive with <code>--recipient</code> .            |
| <code>--armor</code>                           | Armored (base64 PEM) output.                                                                            |
| <code>--pgp-recipient &lt;EMAIL FPR&gt;</code> | PGP recipient (shells out to <code>gpg</code> ).                                                        |

## Examples

```

# Generate
recon --encrypt-keygen > age.key          # public key printed to stderr

# Encrypt
recon --encrypt --recipient age1abc... README.md -o README.md.age
recon --encrypt --passphrase secret.txt -o secret.txt.age --armor

# Decrypt
recon --decrypt --identity age.key README.md.age -o README.md

# PGP
recon --encrypt --pgp-recipient alice@example.com msg.txt -o msg.txt.pgp

```

## Check digits

| Flag                                          | Description                         |
|-----------------------------------------------|-------------------------------------|
| <code>--checkdigit &lt;ALGO&gt;</code>        | Verify an input.                    |
| <code>--checkdigit-create &lt;ALGO&gt;</code> | Compute and append the check digit. |
| <code>--checkdigit-list</code>                | List supported algorithms (70+).    |

Notable algorithms shipped in 0.61.0:

- **Latin-American + Australian + Mexican tax IDs:** `br_cpf`, `br_cnpj`, `ar_cuit`, `ar_cuil`, `cl_rut`, `pe_ruc`, `au_abn`, `mx_rfc`. All except ABN support `--checkdigit-create`; ABN's mod-89 algorithm has no single-digit inverse.
- **110+ year warnings** on Nordic + Bulgarian personal IDs: `cpr` (Denmark), `henkilotunnus` (Finland), `fodselsnummer` (Norway), `bg-egn` (Bulgaria). Same idiom as the Swedish `personnummer`: when the parsed birth year implies age  $\geq 110$ , the verdict's `comment` flags a likely data-entry error.

## Examples

```
recon --checkdigit isbn13 '9780131103627'      # → valid
recon --checkdigit-create isbn13 '978013110362'  # → 9780131103627
recon --checkdigit-create ean13 '400638133393'  # → 4006381333931
recon --checkdigit personnummer '19900101-1239' # Swedish personal ID
recon --checkdigit-list | head -20
```

## Sample data

| Flag                               | Description                          |
|------------------------------------|--------------------------------------|
| <code>--sample &lt;NAME&gt;</code> | Emit a sample payload (faker-style). |
| <code>--sample-list</code>         | List all named samples.              |

## Examples

```
recon --sample-list
recon --sample json-user
recon --sample json-user -o user.json
recon --sample css                # minimal CSS
boilerplate
```

## JWT tokens

| Flag                                      | Description                                                                 |
|-------------------------------------------|-----------------------------------------------------------------------------|
| <code>--jwt-view &lt;TOKEN&gt;</code>     | Decode + pretty-print. No signature check.                                  |
| <code>--jwt-sign &lt;PAYLOAD&gt;</code>   | Sign a JWT. Pair with <code>--jwt-secret</code> or <code>--jwt-key</code> . |
| <code>--jwt-validate &lt;TOKEN&gt;</code> | Validate signature + standard claims.                                       |
| <code>--jwt-secret &lt;STR&gt;</code>     | HMAC secret (HS256/384/512).                                                |
| <code>--jwt-key &lt;PATH&gt;</code>       | RSA/ECDSA/Ed25519 PEM key.                                                  |
| <code>--jwt-alg &lt;ALG&gt;</code>        | HS256, HS384, HS512, RS256, RS384, RS512, ES256, ES384, EdDSA.              |

### Examples

```
recon --jwt-view "$(cat token.txt)"
recon --jwt-sign '{"sub":"alice","exp":9999999999}' --jwt-secret sharedkey
--jwt-alg HS256
recon --jwt-validate "$TOKEN" --jwt-secret sharedkey
recon --jwt-sign '{"sub":"alice"}' --jwt-key private.pem --jwt-alg RS256
```

## Text encoding

| Flag                                 | Description                                |
|--------------------------------------|--------------------------------------------|
| <code>--iconv &lt;SRC:DST&gt;</code> | Standalone <code>iconv</code> replacement. |
| <code>--list-charsets</code>         | List supported charset labels.             |

### Examples

```
recon --iconv utf-8:iso-8859-1 greek.txt -o greek-latin1.txt
echo 'Hello' | recon --iconv utf-8:utf-16le - -o hello.utf16
recon --list-charsets | head -20
```

## Serve mode

| Flag                                                           | Description                                                                  |
|----------------------------------------------------------------|------------------------------------------------------------------------------|
| <code>--serve [ADDR]</code>                                    | Start an HTTP server serving the cwd (default <code>127.0.0.1:8000</code> ). |
| <code>--serve-tls [ADDR]</code>                                | HTTPS. Requires <code>--serve-cert</code> + <code>--serve-key</code> .       |
| <code>--serve-cert &lt;PATH&gt;</code>                         | Server cert (PEM).                                                           |
| <code>--serve-key &lt;PATH&gt;</code>                          | Server key (PEM).                                                            |
| <code>--serve-sni</code><br><code>&lt;HOST:CERT:KEY&gt;</code> | Per-hostname SNI mapping. Repeatable.                                        |

## Examples

```
recon --serve
recon --serve 0.0.0.0:3000
recon --serve-tls 0.0.0.0:8443 --serve-cert cert.pem --serve-key key.pem
recon --serve-tls :443 --serve-sni a.example.com:a.pem:a.key --serve-sni
b.example.com:b.pem:b.key
```

## Write-out format

The `-w '<FORMAT>'` flag prints a curl-compatible summary after the response. Variable names match curl's.

| Variable                           | Description                                |
|------------------------------------|--------------------------------------------|
| <code>%{http_code}</code>          | HTTP status code                           |
| <code>%{size_download}</code>      | Body size in bytes                         |
| <code>%{size_header}</code>        | Response header bytes                      |
| <code>%{size_request}</code>       | Request bytes sent                         |
| <code>%{size_upload}</code>        | Uploaded bytes                             |
| <code>%{speed_download}</code>     | Bytes-per-second                           |
| <code>%{time_total}</code>         | Total operation time (seconds, fractional) |
| <code>%{time_starttransfer}</code> | TTFB                                       |
| <code>%{time_redirect}</code>      | Time spent on redirects                    |
| <code>%{url}</code>                | Final URL (after redirects)                |
| <code>%{url_effective}</code>      | Same as <code>%{url}</code>                |
| <code>%{content_type}</code>       | Response Content-Type                      |

| Variable                                 | Description    |
|------------------------------------------|----------------|
| <code>%{num_redirects}</code>            | Redirect count |
| <code>%{remote_ip}</code>                | Final peer IP  |
| <code>%{remote_port}</code>              | Peer port      |
| <code>%{local_ip} / %{local_port}</code> | Local side     |
| <code>%{scheme}</code>                   | http or https  |

**Not yet accurate:** `time_namelookup`, `time_connect`, `time_appconnect`, `time_pretransfer` render as `0.000000` — see OUT-OF-SCOPE.md.

## Examples

```
recon https://example.com/ -w '\n%{http_code} %{time_total}s %
size_download}B\n'
recon -I https://example.com/ -w '%{scheme}://%{remote_ip}:%{remote_port} %
{http_code}\n'
recon -L https://short.url/x -w '%{num_redirects} hops → %
{url_effective}\n'
```

## Browser automation

| Flag                                          | Description                                                 |
|-----------------------------------------------|-------------------------------------------------------------|
| <code>--browser-screenshot &lt;URL&gt;</code> | Open in agent-browser, save a PNG to <code>-o PATH</code> . |

Requires `agent-browser` on `$PATH`. See [Script engine → agent-browser](#) for deeper automation.

## Examples

```
recon --browser-screenshot https://example.com -o example.png
```

## Meta flags

| Flag                    | Description                                                                 |
|-------------------------|-----------------------------------------------------------------------------|
| <code>--flags</code>    | Alphabetical curl-style flag listing ( <code>--help all</code> equivalent). |
| <code>--examples</code> | Curated examples, paged.                                                    |
| <code>--init</code>     | Bootstrap <code>~/ .recon/</code> (script/, jars/, sni/, config.toml).      |

| Flag                          | Description                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------|
| <code>--editor</code>         | Open response in <code>\$EDITOR</code> .                                                 |
| <code>--editor-cleanup</code> | Remove leftover <code>~/.recon/editor-*</code> tempfiles.                                |
| <code>--no-pager</code>       | Disable paging of <code>--help</code> / <code>--examples</code> / <code>--flags</code> . |

## `--flags` — the quick lookup

`recon --flags` prints every flag alphabetically sorted by long name, curl's `--help all` layout:

```
(short or 4-space pad) --long <VALUE>  short description
```

Short description is capped at ~52 characters (first sentence of each flag's internal help text). Use this as the quick index when you know roughly what you want but not the flag name; follow up with `recon --help <topic>` for the long-form deep-dive.

Example:

```
$ recon --flags | head
  --age                Force the age backend, even if...
  --archive <DEST>    Create an archive
  --armor              Produce ASCII-armored output...
  --bimi <SELECTOR>   Validate the BIMI record
  --browser-screenshot <URL> Open URL in a browser and save...
  --cacert <PATH>     Path to a PEM-encoded CA certificate...
  --cert              Fetch and display the server's TLS cert
  --cert-type <PEM|DER> Format of --client-cert
-E, --client-cert <PATH> Client certificate for mTLS
  --client-key <PATH> Private key for the client certificate
```

The listing is auto-paged through `$PAGER`. Pipe to `grep` for search:

```
recon --flags | grep -i cookie
recon --flags | grep -E '^s*-[a-zA-Z],' # only flags with short keys
```

## Part III — Script engine

---

recon ships an embedded [Rhai](#) interpreter. Every protocol probe, every cryptography primitive, every helper is exposed to scripts. Scripts run via `--script PATH`; `return N` from the top level becomes the process exit code.

### Running scripts

---

```
recon --script ./probe.rhai           # run an absolute /
relative path
recon --script probe                  # falls back to
~/recon/script/probe.rhai
recon --script probe https://example.com alice  # positional args →
args[1], args[2]
```

Script lookup order:

1. Exact path (with or without `.rhai` extension).
2. `~/recon/script/<name>.rhai`.
3. If neither exists — error.

The shipped example scripts in `script/*.rhai` are copy-friendly starting points:

```
cp script/*.rhai ~/recon/script/
recon --script http example.com
```

### Bare-word invocation

After `recon --init` + copying scripts to `~/recon/script/`, scripts become first-class by name:

```
recon --script tcp-echo 127.0.0.1:9000
recon --script doc-convert README.md output.pdf
recon --script client-cert ~/keys/bundle.pem https://mtls.example.com/
```

### Script language (Rhai)

---

Rhai is a lightweight embedded script language. Key facts for newcomers:

- **Let bindings:** `let x = 42;` (mutable by default; use `const` for compile-time constants).
- **Types:** integers (i64), floats (f64), bools, strings, arrays, maps (object literals: `#{ key: value }`), Blob (Vec).
- **String interpolation:** ``Hello ${name}``.
- **Closures / function pointers:** `|a, b| { a + b }`.
- **For loops:** `for i in 0..5 { ... }` (integer ranges), `for item in array { ... }`, `for (k, v) in map { ... }`.
- **Functions:** `fn f(a) { a * 2 }`.
- **Imports:** `import "module-name" as m;` — resolved against the script's directory and `~/.recon/script/`.
- **Error handling:** `try { ... } catch(e) { ... }`.
- **Reserved words** that might surprise: `spawn` (use `thread_spawn`), `async`, `await`, `throw` (all reserved even when unused).

Useful idioms:

```
// Map with computed keys
let cfg = #{
  user: env("USER"),
  host: "api.example.com",
  timeout_ms: 5000,
};

// Chainable array methods
let evens = (0..20).filter(|n| n % 2 == 0).map(|n| n * 10);

// Early return
if !file_exists("/etc/passwd") { return 1; }
```

## Types and literals

Rhai's type system is fixed (no user-defined types at runtime). Recon scripts use the following built-in types:

| Type   | Rust equivalent       | Example literals           |
|--------|-----------------------|----------------------------|
| i64    | 64-bit signed integer | 42, -7, 0xFF, 0b1010, 0o77 |
| f64    | 64-bit float          | 3.14, -1.0e-5, 1_000.5     |
| bool   | Boolean               | true, false                |
| ()     | Unit / null           | ()                         |
| char   | Unicode scalar        | 'a', '\n'                  |
| String | UTF-8 string          | "hello", `hi \${name}`     |

| Type  | Rust equivalent                           | Example literals                       |
|-------|-------------------------------------------|----------------------------------------|
| Array | Heterogeneous dynamic array               | <code>[1, "two", 3.0]</code>           |
| Map   | String-keyed object                       | <code>{ x: 1, y: "ok" }</code>         |
| Blob  | Byte array ( <code>Vec&lt;u8&gt;</code> ) | <code>Blob(), "hello".to_blob()</code> |

```
let n    = 255;           // i64 – integer literal
let f    = 3.14;          // f64
let flag = true;          // bool
let nil  = ();            // unit – Rhai's null
let ch   = 'A';           // char
let s    = "hello";       // string
let tpl  = `n is ${n}`;   // interpolated string (backtick)
let arr  = [1, "two", false]; // heterogeneous array
let obj  = { host: "localhost", port: 8080 }; // map
let raw  = "hello".to_blob(); // Blob of UTF-8 bytes
```

Integer literals accept `_` separators (`1_000_000`) and radix prefixes `0x` (hex), `0b` (binary), `0o` (octal). Type annotations (`let x: i64 = 0;`) are optional — Rhai infers the type.

## Operators

### Arithmetic

| Operator        | Description                    | Example                                               |
|-----------------|--------------------------------|-------------------------------------------------------|
| <code>+</code>  | Addition; string concatenation | <code>1 + 2</code> , <code>"a" + "b"</code>           |
| <code>-</code>  | Subtraction                    | <code>10 - 3</code>                                   |
| <code>*</code>  | Multiplication                 | <code>4 * 5</code>                                    |
| <code>/</code>  | Division (integer truncates)   | <code>7 / 2 → 3</code> ; <code>7.0 / 2.0 → 3.5</code> |
| <code>%</code>  | Remainder                      | <code>10 % 3 → 1</code>                               |
| <code>**</code> | Power                          | <code>2 ** 10 → 1024</code>                           |
| <code>-x</code> | Unary negation                 | <code>-x</code>                                       |

### Comparison (all return `bool`)

`==`, `!=`, `<`, `<=`, `>`, `>=` — work on all numeric types and strings (lexicographic for strings).

### Logical

| Operator                | Description       |
|-------------------------|-------------------|
| <code>&amp;&amp;</code> | Short-circuit AND |
| <code>  </code>         | Short-circuit OR  |
| <code>!</code>          | Logical NOT       |

**Bitwise** (integers only)

`&` (AND), `|` (OR), `^` (XOR), `~` (bitwise NOT), `<<` (left shift), `>>` (right shift).

**String concatenation**

`"hello " + "world"` — either operand may be a non-string and is automatically converted:  
`"n=" + 42 → "n=42"`.

**Null-coalescing**

`value ?? default` evaluates to `default` when `value` is `()`, otherwise `value`. Useful for optional map lookups — accessing a missing key returns `()`:

```
let port    = opts["port"] ?? 80;
let charset = response["content-type"] ?? "application/octet-stream";
```

Note: `env()` always returns a `String` (empty string when unset), never `()`, so `??` does not apply to it. Use the two-argument form for env defaults: `env("HOME", "/tmp")`.

**Ranges**

`1..5` (exclusive — does not include 5), `1..=5` (inclusive — includes 5). Ranges are iterable and support `.filter()`, `.map()`, `.reduce()`.

**Combined example**

```
let x = 7;
let y = 2;
print(x + y);           // 9
print(x ** y);          // 49
print(x % y);           // 1
print(x > 5 && y < 10); // true
print((x | y) == 7);    // true (0b111 | 0b010 = 0b111)
let label = env("LABEL", "default"); // env() never returns () — use 2-arg form
print(`label is ${label}`);
let sum = (1..=10).reduce(|acc, n| acc + n, 0);
print(sum);             // 55
```

## Control flow

### if / else if / else

```
if x > 10 {  
    print("big");  
} else if x > 0 {  
    print("small");  
} else {  
    print("non-positive");  
}
```

If blocks are expressions and return the last evaluated value:

```
let label = if x > 0 { "positive" } else { "non-positive" };
```

### Ternary operator

```
let sign = x > 0 ? "+" : "-";
```

### while

```
let i = 0;  
while i < 5 {  
    print(i);  
    i += 1;  
}
```

### loop

`loop { ... }` runs until `break`:

```
let attempts = 0;  
loop {  
    attempts += 1;  
    if attempts >= 3 { break; }  
}
```

### for — range

```
for i in 0..5 { print(i); } // 0 1 2 3 4 (exclusive)  
for i in 1..=5 { print(i); } // 1 2 3 4 5 (inclusive)
```

### for — array

```
let hosts = ["a.com", "b.com", "c.com"];
for host in hosts {
    print(host);
}
```

### for — map

```
let cfg = #{ host: "a.com", port: 443 };
for (key, val) in cfg {
    print(`${key} = ${val}`);
}
```

### break / continue

```
for i in 0..10 {
    if i == 3 { continue; } // skip 3
    if i == 7 { break; }    // stop before 7
    print(i);
}
```

### return

`return` exits the current function (or the script when used at top level):

```
fn first_positive(arr) {
    for n in arr {
        if n > 0 { return n; }
    }
    () // not found – implicit unit return
}
```

---

## Functions and closures

### Named functions

```
fn add(a, b) { a + b }           // last expression is implicit return value

fn factorial(n) {
    if n <= 1 { return 1; }
    n * factorial(n - 1)         // recursion works
}

print(add(3, 4));                // 7
print(factorial(5));             // 120
```

Rhai hoists function definitions, so a function can be called before it appears in the file.

## Closures

```
let double = |x| x * 2;
let greet  = |name| `Hello, ${name}!`;

print(double(5));      // 10
print(greet("world")); // Hello, world!

let nums  = [1, 2, 3, 4, 5];
let evens = nums.filter(|n| n % 2 == 0); // [2, 4]
let square = evens.map(|n| n * n);      // [4, 16]
print(square);
```

Closures capture outer variables by value at creation time.

## Function pointers

```
fn triple(n) { n * 3 }

let fp = Fn("triple"); // function pointer by name
print(call(fp, 7));    // 21
```

---

## Error handling

### try / catch

In Rhai 1.x, `e` in a `catch` block is the error message as a string:

```
try {
    let r = http("https://httpbin.org/status/500");
    if r.status != 200 {
        throw `HTTP error: ${r.status}`;
    }
    print(r.body);
} catch(e) {
    eprint(`Error: ${e}`);
}
```

### throw

`throw "message"` raises an error that propagates to the nearest `catch` block or terminates the script with a non-zero exit code:

```
fn parse_port(s) {
  let n = s.to_int();
  if n < 1 || n > 65535 { throw `invalid port: ${s}`; }
  n
}
```

## assert

`assert(cond, msg)` is a recon helper that calls `throw msg` when `cond` is `false`. Prefer it over manual `if/throw` when validating invariants:

```
let r = http("https://httpbin.org/get");
assert(r.status == 200, `Expected 200, got ${r.status}`);
```

## Standard built-in functions

The following are available in every recon script without any import. They come from Rhai's standard packages, augmented by recon's own helpers.

### Type inspection and conversion

| Expression                  | Description                | Example result                                                  |
|-----------------------------|----------------------------|-----------------------------------------------------------------|
| <code>type_of(v)</code>     | Runtime type name          | "i64", "f64", "string", "array",<br>"map", "bool", "()", "blob" |
| <code>v.to_int()</code>     | Convert to i64             | "42".to_int() → 42                                              |
| <code>v.to_float()</code>   | Convert to f64             | 3.to_float() → 3.0                                              |
| <code>v.to_string()</code>  | Convert to String          | true.to_string() → "true"                                       |
| <code>parse_int(s)</code>   | Parse decimal string → i64 | parse_int("255") → 255                                          |
| <code>parse_float(s)</code> | Parse string → f64         | parse_float("3.14") → 3.14                                      |

```
let v = "123".to_int();
print(type_of(v));           // "i64"
print(type_of(3.14));        // "f64"
print(type_of([1, 2]));      // "array"
print(type_of(()));          // "()"
print(42.to_float() + 0.5);  // 42.5
print(parse_int("0xFF"));    // 255
```

## String methods

Strings in Rhai are immutable. Methods that appear to "modify" a string return a new one.

| Method                                                    | Description                                             |
|-----------------------------------------------------------|---------------------------------------------------------|
| <code>.len()</code>                                       | Character (code-point) count                            |
| <code>.is_empty()</code>                                  | <code>true</code> if <code>""</code>                    |
| <code>.to_upper()</code>                                  | Uppercase copy                                          |
| <code>.to_lower()</code>                                  | Lowercase copy                                          |
| <code>.trim()</code>                                      | Strip leading and trailing whitespace                   |
| <code>.trim_start()</code> / <code>.trim_end()</code>     | Strip one side only                                     |
| <code>.contains(sub)</code>                               | Substring check; returns <code>bool</code>              |
| <code>.starts_with(s)</code> / <code>.ends_with(s)</code> | Prefix / suffix check                                   |
| <code>.replace(old, new)</code>                           | Replace all occurrences                                 |
| <code>.split(sep)</code>                                  | Split on separator; returns array                       |
| <code>.chars()</code>                                     | Array of single-character strings                       |
| <code>.sub_string(start)</code>                           | Slice from index to end                                 |
| <code>.sub_string(start, len)</code>                      | Slice of given length                                   |
| <code>.pad(len, ch)</code>                                | Right-pad to <code>len</code> with char <code>ch</code> |
| <code>.index_of(sub)</code>                               | First match index or <code>-1</code>                    |
| <code>.to_blob()</code>                                   | Encode as UTF-8 <code>Blob</code>                       |
| <code>.to_int()</code>                                    | Parse as decimal integer                                |
| <code>.to_float()</code>                                  | Parse as float                                          |

```
let raw = " https://Example.COM/Path ";
let url = raw.trim().to_lower();           // "https://example.com/path"
print(url.contains("example"));           // true
print(url.starts_with("https"));          // true
print(url.replace("example.com", "cdn.example.com"));

let parts = "a,b,c,d".split(",");         // ["a", "b", "c", "d"]
print(parts.len());                       // 4
print(parts.join(" | "));                 // "a | b | c | d"

let host = "api.example.com";
print(host.sub_string(4, 7));              // "example"
print(host.index_of("."));                 // 3
```

## Array methods

| Method                         | Description                                                            |
|--------------------------------|------------------------------------------------------------------------|
| <code>.len()</code>            | Number of elements                                                     |
| <code>.is_empty()</code>       | <code>true</code> if array is empty                                    |
| <code>.push(v)</code>          | Append element (in-place)                                              |
| <code>.pop()</code>            | Remove and return last element                                         |
| <code>.shift()</code>          | Remove and return first element                                        |
| <code>.unshift(v)</code>       | Prepend element (in-place)                                             |
| <code>.insert(i, v)</code>     | Insert at index <code>i</code> (in-place)                              |
| <code>.remove(i)</code>        | Remove element at index <code>i</code> and return it                   |
| <code>.reverse()</code>        | Reverse in-place                                                       |
| <code>.contains(v)</code>      | Membership test                                                        |
| <code>.index_of(v)</code>      | First index of <code>v</code> , or <code>-1</code>                     |
| <code>.join(sep)</code>        | Concatenate elements as strings with separator                         |
| <code>.map(fn)</code>          | Apply function; returns new array                                      |
| <code>.filter(fn)</code>       | Keep elements where predicate is <code>true</code> ; returns new array |
| <code>.reduce(fn, init)</code> | Fold to single value                                                   |
| <code>.sort()</code>           | In-place ascending sort                                                |
| <code>.sort(fn)</code>         | In-place sort with comparator <code>fn(a, b) -&gt; i64</code>          |
| <code>.dedup()</code>          | Remove consecutive duplicate elements (in-place)                       |
| <code>.drain(range)</code>     | Remove and return elements in index range                              |

```

let nums = [5, 3, 8, 1, 9, 2];
nums.sort();                                // [1, 2, 3, 5, 8, 9]

let big  = nums.filter(|n| n > 4);           // [5, 8, 9]
let dbl  = big.map(|n| n * 2);               // [10, 16, 18]
let sum  = dbl.reduce(|acc, n| acc + n, 0);   // 44

nums.push(42);
nums.unshift(0);
print(nums.len());                          // 8
print(nums.contains(42));                   // true
print(nums.index_of(3));                    // 2 (after sort)

// Negative indices index from the end
print(nums[-1]);                           // last element

```

## Map methods

Maps use string keys. Literal syntax is `#{ key: value }`. Keys with special characters or spaces must be quoted: `#{ "content-type": "json" }`.

| Method / accessor                     | Description                     |
|---------------------------------------|---------------------------------|
| <code>.len()</code>                   | Number of key-value pairs       |
| <code>.is_empty()</code>              | <code>true</code> if no entries |
| <code>.keys()</code>                  | Array of all key strings        |
| <code>.values()</code>                | Array of all values             |
| <code>.contains_key(k)</code>         | Key existence check             |
| <code>.remove(k)</code>               | Delete key and return its value |
| <code>m.key / m["key"]</code>         | Read a value                    |
| <code>m.key = v / m["key"] = v</code> | Write a value                   |

```
let hdr = #{
  "content-type": "application/json",
  accept: "application/json",
};
print(hdr.len());           // 2
print(hdr.keys());          // ["content-type", "accept"]
print(hdr.contains_key("accept")); // true

hdr["x-request-id"] = "abc-123";

for (k, v) in hdr {
  print(`${k}: ${v}`);
}

let ct = hdr.remove("content-type");
print(ct);                  // "application/json"
print(hdr.len());           // 2
```

## Math functions

| Function                           | Description                                    |
|------------------------------------|------------------------------------------------|
| <code>abs(n)</code>                | Absolute value                                 |
| <code>sqrt(n)</code>               | Square root (returns <code>f64</code> )        |
| <code>pow(base, exp)</code>        | Power — equivalent to <code>base ** exp</code> |
| <code>min(a, b) / max(a, b)</code> | Minimum / maximum                              |
| <code>round(n)</code>              | Round to nearest integer                       |

| Function                                                           | Description                               |
|--------------------------------------------------------------------|-------------------------------------------|
| <code>floor(n)</code> / <code>ceil(n)</code>                       | Floor / ceiling                           |
| <code>sin(n)</code> / <code>cos(n)</code> / <code>tan(n)</code>    | Trig functions (radians)                  |
| <code>asin(n)</code> / <code>acos(n)</code> / <code>atan(n)</code> | Inverse trig (radians)                    |
| <code>ln(n)</code>                                                 | Natural logarithm (base e)                |
| <code>log(n)</code>                                                | Base-10 logarithm                         |
| <code>exp(n)</code>                                                | $e^n$                                     |
| PI                                                                 | $\pi \approx 3.14159265358979$            |
| E                                                                  | Euler's number $\approx 2.71828182845905$ |

```

print(abs(-7));           // 7
print(sqrt(144.0));       // 12.0
print(pow(2, 10));        // 1024
print(round(3.7));        // 4
print(floor(3.7));        // 3
print(ceil(3.1));         // 4
print(min(5, 3));         // 3
print(max(5, 3));         // 5

let angle = PI / 4.0;
print(sin(angle));        // ≈ 0.7071 (sin 45°)
print(exp(1.0));          // ≈ 2.7183 (same as E)

```

## Ranges

Ranges are lazy integer sequences.

| Expression         | Meaning                                                                       |
|--------------------|-------------------------------------------------------------------------------|
| <code>a..b</code>  | Exclusive: integers <code>a</code> , <code>a+1</code> , ..., <code>b-1</code> |
| <code>a..=b</code> | Inclusive: integers <code>a</code> , <code>a+1</code> , ..., <code>b</code>   |

Ranges support `.filter()`, `.map()`, `.reduce()`, `.for_each()`. A range can also be used as an array slice index: `arr[start..end]`.

```
// Sum 1 to 100
let total = (1..=100).reduce(|acc, n| acc + n, 0);
print(total);    // 5050

// Collect even squares
let evens = (1..=10)
    .filter(|n| n % 2 == 0)
    .map(|n| n * n);
print(evens);    // [4, 16, 36, 64, 100]

// Array slice with range
let letters = ["a", "b", "c", "d", "e"];
let mid = letters[1..4];    // ["b", "c", "d"]
print(mid);
```

---

## Shebang — executable scripts

A `.rhai` file can be made directly executable by adding a shebang as the first line:

```
#!/usr/bin/env -S recon --script
```

The `-S` flag instructs `/usr/bin/env` to split its argument on whitespace, so `recon` receives `--script` as a separate flag. This works on macOS and all major Linux distributions.

### Full example

```
cat > ~/bin/health <<'EOF'
#!/usr/bin/env -S recon --script
let host = if args.len() > 1 { args[1] } else { "example.com" };
let r = https(`https://${host}`);
print(`${r.status} ${host} (${r.duration_ms}ms)`);
return if r.status == 200 { 0 } else { 1 };
EOF
chmod +x ~/bin/health

./health example.com          # run directly
./health api.example.com      # host from args[1]
```

### How it works

When the kernel executes a shebang file, it calls the interpreter with the script path as an argument — equivalent to `recon --script ./health`. Recon detects a leading `#!` in the source and replaces it with `//` (a Rhai line comment) before compilation. This preserves line numbers in error messages while making the file valid Rhai.

## Arguments and flags

Trailing arguments after the script name land in `args[1..]` exactly as with `--script`. CLI flags (`-k`, `-v`, `--connect-timeout`, etc.) set before or after the script name in the invocation are reflected in the `flags` map inside the script.

```
./health api.example.com          # args[1] = "api.example.com"
recon -k ./health api.example.com # flags.insecure = true inside script
```

## Notes

- On older systems where `/usr/bin/env -S` is not available, use the full path:  
`#!/usr/local/bin/recon --script` (hard-coded path, no `-S` needed).
- The shebang is only stripped when it appears on the **first line**. A `#!` appearing anywhere else in the file remains a parse error (Rhai does not treat `#` as a comment character).
- `recon --help shebang` summarises the shebang feature.

## CLI inheritance

When a script runs, two read-only constants land at the top of its scope:

- **args** — an array of strings. `args[0]` is the script path; `args[1..]` are the positional arguments after `--script PATH`.
- **flags** — a map mirroring the relevant CLI flags (lower-case, snake\_case keys). Useful for scripts that want to respect `-v`, `-k`, `--connect-timeout`, etc. at invocation time.

```
// Reasonable defaults, overridable via args
let url = if args.len() > 1 { args[1] } else { "https://example.com/" };
let timeout = flags.connect_timeout; // from -C / --connect-timeout

let r = http(url, #{ timeout_ms: timeout * 1000 });
print(`${r.status} ${r.url}`);
```

Bindings that take their own opts map (e.g. `http(url, opts)`) layer the caller's opts ON TOP of the CLI defaults. If the caller wants to ignore the CLI's `-H` header list, they can pass `headers: []` in their opts.

## Core helpers

Exposed by `src/script/bindings/helpers.rs`. Always available.

| Function                               | Returns | Description                                                                   |
|----------------------------------------|---------|-------------------------------------------------------------------------------|
| <code>sleep_ms(ms)</code>              | —       | Block the current thread.                                                     |
| <code>sleep(ms)</code>                 | —       | Alias of <code>sleep_ms</code> (added in 0.56.0 for thread-side readability). |
| <code>env(name)</code>                 | string  | Process environment variable; <code>""</code> if unset.                       |
| <code>env(name, default)</code>        | string  | With fallback.                                                                |
| <code>now()</code>                     | int     | Unix seconds.                                                                 |
| <code>now_ms()</code>                  | int     | Unix milliseconds.                                                            |
| <code>assert(cond, msg)</code>         | —       | Throws on false.                                                              |
| <code>json_parse(s)</code>             | Dynamic | Parse JSON text → native Rhai value.                                          |
| <code>json_stringify(v)</code>         | string  | Serialize to compact JSON.                                                    |
| <code>json_stringify(v, pretty)</code> | string  | <code>pretty: true</code> = 2-space indent.                                   |
| <code>json_stringify(v, n)</code>      | string  | <code>n</code> = spaces per indent (clamped 1..=8).                           |

Rhai's built-in `print(x)` writes `x` + newline via the engine's debug callback. `recon` adds byte-precise siblings (see next section).

## Examples

```

// Probe a URL, fail fast
let r = http("https://api.example.com/health");
assert(r.status == 200, `expected 200, got ${r.status}`);

// JSON round-trip
let obj = json_parse(r.body);
print(json_stringify(obj, true));    // pretty

// Time gate
let t0 = now_ms();
let r = http("https://slow.example.com/");
let dt = now_ms() - t0;
print(`took ${dt} ms`);

// Polling with timeout
let deadline = now() + 30;
while now() < deadline {
    let r = http("https://example.com/ready");
    if r.status == 200 { return 0; }
    sleep_ms(500);
}
return 1;

```

## Output bindings

0.53.0. Byte-precise output that Rhai's built-in `print()` can't do.

| Function                                                   | Description                                        |
|------------------------------------------------------------|----------------------------------------------------|
| <code>print_raw(s)</code> / <code>print_raw(blob)</code>   | Write to stdout without a trailing newline; flush. |
| <code>eprint(s)</code>                                     | Write to stderr with a newline.                    |
| <code>eprint_raw(s)</code> / <code>eprint_raw(blob)</code> | Write to stderr without newline; flush.            |
| <code>flush()</code>                                       | Explicit stdout flush.                             |

## Examples

```
// Progress indicator
print_raw("loading");
for i in 0..10 {
    sleep_ms(200);
    print_raw(".");
}
print_raw("\n");

// Send bytes to stdout (e.g. for piping into another tool)
let bytes = http("https://example.com/image.png").body;
print_raw(bytes);

// Log to stderr; keep stdout for data
eprint(`[${now_ms()}] starting probe...`);
print_raw(json_stringify(result));    // stdout stays parseable
```

## File I/O bindings

0.53.0. Both whole-file convenience and a streaming handle API.

### Whole-file helpers

| Function                                     | Description                       |
|----------------------------------------------|-----------------------------------|
| <code>file_read(path)</code>                 | Read entire file → Blob.          |
| <code>file_write_all(path, blob str)</code>  | Overwrite. Returns bytes written. |
| <code>file_append_all(path, blob str)</code> | Append; creates if absent.        |
| <code>file_exists(path)</code>               | Bool.                             |
| <code>file_size(path)</code>                 | Bytes.                            |
| <code>file_delete(path)</code>               | Unlink.                           |

`path` accepts a filesystem path or a `file://` URL.

### Streaming handle API

| Function                           | Description                                                                                                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>file_open(path, mode)</code> | Returns a <code>FileHandle</code> . Modes: <code>r</code> , <code>w</code> (truncate+create), <code>rw</code> , <code>rw+</code> / <code>w+</code> (read+write+create+truncate), <code>a</code> (append+create), <code>ra</code> (append+read). |
| <code>file_read(h, n)</code>       | Read up to <code>n</code> bytes → Blob.                                                                                                                                                                                                         |
| <code>file_read_all(h)</code>      | Drain to Blob.                                                                                                                                                                                                                                  |

| Function                               | Description                                                                               |
|----------------------------------------|-------------------------------------------------------------------------------------------|
| <code>file_write(h, blob str)</code>   | Write all bytes. Returns count.                                                           |
| <code>file_seek(h, pos, whence)</code> | whence = <code>start</code>   <code>cur</code>   <code>end</code> . Returns new position. |
| <code>file_tell(h)</code>              | Current position.                                                                         |
| <code>file_flush(h)</code>             | Sync buffered writes.                                                                     |
| <code>file_close(h)</code>             | Close (drops the underlying file).                                                        |

## Examples

```
// Whole-file
let data = file_read("config.json");
file_write_all("/tmp/copy.json", data);
file_append_all("/tmp/audit.log", `probe at ${now()}\n`);

// Check before read
if !file_exists("secrets.key") {
    eprint("missing key");
    return 2;
}

// Streaming – copy with 4KB chunks
let src = file_open("big.bin", "r");
let dst = file_open("/tmp/big.bin.copy", "w");
loop {
    let chunk = file_read(src, 4096);
    if chunk.len() == 0 { break; }
    file_write(dst, chunk);
}
file_close(src);
file_close(dst);

// Random access
let h = file_open("/tmp/log.bin", "rwc");
file_write(h, "record-1\n");
file_write(h, "record-2\n");
file_seek(h, 0, "start");
let head = file_read_all(h);
print(`contents: ${head.len()} bytes`);
file_close(h);
```

# Comparison bindings

0.53.0. Diff two in-memory values (strings or Blobs).

| Function                   | Returns | Description                                                                  |
|----------------------------|---------|------------------------------------------------------------------------------|
| <code>compare(a, b)</code> | Map     | <code>{ identical, binary, a_bytes, b_bytes, added, removed, diff }</code> . |

Both `a` and `b` accept strings OR Blobs. The binding does NOT fetch URLs — scripts should pre-load via `http()` or `file_read()` and pass the bytes.

## Examples

```
// Compare two API responses
let a = http("https://api.v1.example.com/users/42").body;
let b = http("https://api.v2.example.com/users/42").body;
let r = compare(a, b);
if r.identical {
    print("parity OK");
    return 0;
}
print(`diverged: +${r.added} / -${r.removed}`);
print_raw(r.diff);
return 1;

// Compare a live URL against a baseline file
let live = http("https://example.com/status").body;
let baseline = file_read("status.baseline").to_string();
let r = compare(live.to_string(), baseline);
if !r.identical { print_raw(r.diff); }

// Binary content
let old = file_read("logo.v1.png");
let new = file_read("logo.v2.png");
let r = compare(old, new);
print(`binary: ${r.binary}, size delta: ${r.b_bytes - r.a_bytes}`);
```

## HTTP binding

The workhorse. Two call forms:

```
http(url)           → response map
http(url, opts)     → response map
```

## Response map

| Key          | Type          | Description                                                           |
|--------------|---------------|-----------------------------------------------------------------------|
| status       | int           | HTTP status code.                                                     |
| url          | string        | Original URL.                                                         |
| final_url    | string        | After redirects.                                                      |
| headers      | Map           | Response headers; values are strings (first) or arrays when repeated. |
| body         | Blob   string | Response body. String when content-type is text-ish; Blob otherwise.  |
| duration_ms  | int           | Total wall-clock time.                                                |
| http_version | string        | HTTP/1.1 , HTTP/2 , etc.                                              |

## Options map

Every key is optional.

| Key              | Type          | Description                                                                                                                             |
|------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| method           | string        | GET, POST, PUT, PATCH, DELETE, HEAD.                                                                                                    |
| headers          | Map           | Key → value (or array of values). Merged with CLI <code>-H</code> defaults.                                                             |
| body             | string / Blob | Request body.                                                                                                                           |
| json             | Dynamic       | Serialize as JSON, set <code>Content-Type</code> + <code>Accept</code> .                                                                |
| form             | Map           | URL-encoded form body.                                                                                                                  |
| multipart        | Array of Maps | Each Map has <code>name</code> , <code>value</code> or <code>file</code> , optional <code>filename</code> , <code>content_type</code> . |
| basic_auth       | string        | <code>"user:pass"</code> .                                                                                                              |
| bearer           | string        | Bearer token.                                                                                                                           |
| user_agent       | string        | Override User-Agent.                                                                                                                    |
| referer          | string        | Referer.                                                                                                                                |
| timeout_ms       | int           | Total operation timeout.                                                                                                                |
| connect_timeout  | int           | TCP connect timeout (seconds).                                                                                                          |
| insecure         | bool          | Skip TLS verification.                                                                                                                  |
| follow_redirects | bool          | Enable / disable redirect follow.                                                                                                       |

| Key                                   | Type   | Description                                                                                |
|---------------------------------------|--------|--------------------------------------------------------------------------------------------|
| <code>max_redirects</code>            | int    | Redirect cap.                                                                              |
| <code>compressed</code>               | bool   | Request + auto-decompress.                                                                 |
| <code>cookiejar</code>                | string | Path to a SQLite cookie jar.                                                               |
| <code>proxy</code>                    | string | Proxy URL.                                                                                 |
| <code>proxy_user</code>               | string | Proxy basic auth.                                                                          |
| <code>noproxy</code>                  | string | Bypass list.                                                                               |
| <code>proxy_insecure</code>           | bool   | Skip cert verification on proxy.                                                           |
| <code>proxy_cacert</code>             | string | Extra CA for proxy.                                                                        |
| <code>unix_socket</code>              | string | Route via Unix socket.                                                                     |
| <code>hsts</code>                     | string | Path to HSTS cache.                                                                        |
| <code>client_cert</code>              | string | PEM client cert.                                                                           |
| <code>client_key</code>               | string | PEM client key.                                                                            |
| <code>cert_type /<br/>key_type</code> | string | PEM   DER   ENG (see CLI docs).                                                            |
| <code>pass</code>                     | string | PKCS#8 passphrase (reserved).                                                              |
| <code>wait</code>                     | int    | (0.67.0) Seconds between URLs in batch mode (CLI-only effect).                             |
| <code>tries</code>                    | int    | (0.67.0) Total attempts; overrides <code>retry</code> . <code>tries = retries + 1</code> . |
| <code>accept</code>                   | string | (0.67.0) Comma-separated filename-suffix accept list.                                      |
| <code>reject</code>                   | string | (0.67.0) Comma-separated filename-suffix reject list.                                      |

## Examples

```

// Simple GET
let r = http("https://api.example.com/status");
print(`${r.status} ${r.body.len()} bytes`);

// POST JSON
let r = http("https://api.example.com/users", #{
  method: "POST",
  json: #{ name: "alice", email: "a@example.com" },
});
assert(r.status == 201, "create failed");

// Form POST
http("https://api.example.com/login", #{
  method: "POST",
  form: #{ user: "alice", pass: "s3cr3t" },
  cookiejar: "/tmp/jar.db",
});

// Multipart upload
http("https://upload.example.com/avatars", #{
  method: "POST",
  multipart: [
    #{ name: "title", value: "profile" },
    #{ name: "image", file: "avatar.png", content_type: "image/png" },
  ],
  bearer: env("API_TOKEN"),
});

// Bearer auth + timeout + compressed
let r = http("https://api.example.com/items", #{
  bearer: env("API_TOKEN"),
  timeout_ms: 5000,
  compressed: true,
  headers: #{ "Accept": "application/json" },
});

// Follow + retry on 5xx
for attempt in 0..3 {
  let r = http("https://api.example.com/", #{ follow_redirects: true });
  if r.status < 500 { return r.status == 200 ? 0 : 1; }
  sleep_ms(1000 * (attempt + 1));
}
return 2;

// Proxy + HSTS
http("http://example.com/", #{
  proxy: "socks5h://127.0.0.1:9050",
  hsts: "/tmp/hsts.txt",
});

```

```
});

// mTLS
http("https://mtls.example.com/", #{
  client_cert: "/etc/keys/bundle.pem",
});

// Unix socket
let r = http("http://localhost/v1.40/version", #{
  unix_socket: "/var/run/docker.sock",
});
print(json_stringify(json_parse(r.body), 2));
```

## Browser (sticky-session) binding

A stateful HTTP "browser" — cookies and headers persist across calls against the same handle. Matches the common "log in once, then make authenticated requests" pattern.

| Function                                                                     | Description                                                         |
|------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <code>browser()</code>                                                       | New browser with a tempfile cookie jar.                             |
| <code>browser(opts)</code>                                                   | With initial config.                                                |
| <code>use_persistent_session(name)</code>                                    | Browser method; jar is <code>~/.recon/jars/&lt;name&gt;.db</code> . |
| <code>&lt;br&gt;.get(url [, opts])</code>                                    | GET via the browser.                                                |
| <code>&lt;br&gt;.post(url, opts)</code>                                      | POST.                                                               |
| <code>&lt;br&gt;.put(url, opts) / .patch(...) / .delete(url [, opts])</code> | Other methods.                                                      |
| <code>&lt;br&gt;.request(method, url, opts)</code>                           | Catch-all.                                                          |
| <code>&lt;br&gt;.header(name, value)</code>                                  | Sticky request header.                                              |
| <code>&lt;br&gt;.headers(map)</code>                                         | Merge sticky headers.                                               |
| <code>&lt;br&gt;.clear_headers()</code>                                      | Remove sticky headers.                                              |
| <code>&lt;br&gt;.user_agent(str)</code>                                      | Override User-Agent for the lifetime of the browser.                |
| <code>&lt;br&gt;.basic_auth(user, pass)</code>                               | Sticky Basic auth.                                                  |
| <code>&lt;br&gt;.timeout_ms(n) / .connect_timeout(secs)</code>               | Sticky timeouts.                                                    |

| Function                                                                                     | Description                           |
|----------------------------------------------------------------------------------------------|---------------------------------------|
| <code>&lt;br&gt;.insecure(bool) /<br/>.follow_redirects(bool) /<br/>.max_redirects(n)</code> | Sticky knobs.                         |
| <code>&lt;br&gt;.cookiejar()</code>                                                          | Current jar path.                     |
| <code>&lt;br&gt;.cookies()</code>                                                            | Array of cookie records from the jar. |
| <code>&lt;br&gt;.cookie_set(name, value, domain,<br/>path)</code>                            | Inject a cookie.                      |

## Examples

```
// Login, then fetch protected resources
let br = browser();
br.user_agent("recon-test/0.58");
br.post("https://example.com/login", #{
  form: #{ user: "alice", pass: "s3cr3t" },
});
let r = br.get("https://example.com/dashboard");
assert(r.status == 200, "dashboard fetch failed");

// Persist across runs – next invocation sees the same cookies
let br = browser();
br.use_persistent_session("gh-session");
br.header("Accept", "application/vnd.github+json");
let r = br.get("https://api.github.com/user");

// Multi-persona fan-out
let personas = [{ name: "a", jar: "persona-a" }, { name: "b", jar:
"persona-b" }];
for p in personas {
  let br = browser();
  br.use_persistent_session(p.jar);
  let r = br.get("https://api.example.com/whoami");
  print(`${p.name}: ${r.status}`);
}
```

## TCP, TCP-server, UDP

0.57.0 shipped the server bindings. Handles are Send+Sync so they survive `thread_spawn` (0.56.0) boundaries — the expected pattern is "accept on main, spawn a handler per connection".

## TCP client (existing)

| Function                    | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <code>tcp(url)</code>       | TCP connect probe; returns <code>{ status, duration_ms, ... }</code> . |
| <code>tcp(url, opts)</code> | With <code>timeout</code> (ms).                                        |

## TCP server (0.57.0)

| Function                                      | Description                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------------------|
| <code>tcp_listen(addr)</code>                 | Bind. <code>addr</code> like <code>"0.0.0.0:8080"</code> or <code>"[:]:8080"</code> . |
| <code>tcp_accept(listener)</code>             | Blocking accept. Returns a <code>TcpConn</code> .                                     |
| <code>tcp_accept(listener, timeout_ms)</code> | Times out with an error.                                                              |
| <code>tcp_read(conn, n, timeout_ms)</code>    | Read up to N bytes → Blob.                                                            |
| <code>tcp_read_line(conn, timeout_ms)</code>  | <code>\n</code> -terminated line; CR/LF stripped.                                     |
| <code>tcp_write(conn, blob str)</code>        | Write all. Returns bytes.                                                             |
| <code>tcp_peer_addr(conn)</code>              | Remote <code>SocketAddr</code> string.                                                |
| <code>tcp_close(conn)</code>                  | Close connection.                                                                     |
| <code>tcp_close_listener(l)</code>            | Close listener.                                                                       |

## UDP

| Function                                              | Description                                                                         |
|-------------------------------------------------------|-------------------------------------------------------------------------------------|
| <code>udp_bind(addr)</code>                           | Bind.                                                                               |
| <code>udp_recv_from(sock, max_len)</code>             | Block until a datagram arrives. Returns <code>{ data: Blob, addr: string }</code> . |
| <code>udp_recv_from(sock, max_len, timeout_ms)</code> | With timeout.                                                                       |
| <code>udp_send_to(sock, blob str, addr)</code>        | Returns bytes sent.                                                                 |
| <code>udp_close(sock)</code>                          | Release.                                                                            |

## Examples

```
// TCP client probe
let r = tcp("example.com:443");
print(`${r.status} in ${r.duration_ms}ms`);

// Concurrent TCP echo server (from script/tcp-echo.rhai)
let l = tcp_listen("127.0.0.1:9000");
loop {
    let conn = tcp_accept(l);
    thread_spawn(|c| {
        let peer = tcp_peer_addr(c);
        let line = tcp_read_line(c, 5000);
        tcp_write(c, `echo: ${line}` + "\n");
        tcp_close(c);
    }, conn);
}

// UDP listener (from script/udp-listen.rhai)
let s = udp_bind("127.0.0.1:9001");
loop {
    let r = udp_recv_from(s, 65536, 30000);
    print(`${r.addr} → ${r.data.len()} bytes`);
}

// UDP beacon sender
let s = udp_bind("0.0.0.0:0");
for i in 0..5 {
    udp_send_to(s, `beacon-${i}`, "127.0.0.1:9001");
    sleep_ms(200);
}
udp_close(s);
```

## Threading bindings

0.56.0. Requires rhai's `sync` feature (enabled). `spawn` alone is reserved by Rhai — use `thread_spawn`.

| Function                                      | Description                                                        |
|-----------------------------------------------|--------------------------------------------------------------------|
| <code>thread_spawn(fn_ptr)</code>             | Spawn a closure. Returns a <code>ThreadHandle</code> .             |
| <code>thread_spawn(fn_ptr, arg)</code>        | With one forwarded arg.                                            |
| <code>thread_spawn(fn_ptr, args_array)</code> | With N args.                                                       |
| <code>join(h)</code>                          | Block; returns the closure's return value (or the worker's error). |

| Function                          | Description                                                |
|-----------------------------------|------------------------------------------------------------|
| <code>tid()</code>                | Current thread id (stable within a run).                   |
| <code>sleep(ms)</code>            | Alias of <code>sleep_ms</code> .                           |
| <code>channel()</code>            | Unbounded MPSC — returns <code>[sender, receiver]</code> . |
| <code>channel_bounded(n)</code>   | Bounded; <code>try_send</code> returns false when full.    |
| <code>send(tx, val)</code>        | Blocking send.                                             |
| <code>try_send(tx, val)</code>    | Non-blocking; false on full.                               |
| <code>recv(rx)</code>             | Blocking.                                                  |
| <code>recv(rx, timeout_ms)</code> | With timeout.                                              |
| <code>try_recv(rx)</code>         | Non-blocking; <code>()</code> when empty.                  |

## Examples

```

// Fan out probes, gather via channel
let c = channel();
let tx = c[0];
let rx = c[1];

let urls = [
    "https://a.example.com/",
    "https://b.example.com/",
    "https://c.example.com/",
];

for url in urls {
    thread_spawn(|u| {
        let r = http(u, #{ timeout_ms: 5000 });
        send(tx, `${u} → ${r.status}`);
    }, url);
}

for i in 0..urls.len() {
    print(recv(rx, 10000));
}

// Bounded channel – producer/consumer with back-pressure
let c = channel_bounded(4);
let tx = c[0];
let rx = c[1];

thread_spawn(|| {
    for i in 0..100 { send(tx, i); }
});

let sum = 0;
for j in 0..100 {
    sum += recv(rx, 1000);
}
print(`total: ${sum}`);

// Join for return value
let h = thread_spawn(|| {
    // Do something heavy
    let r = http("https://slow.example.com/");
    r.status
});
let status = join(h);
print(`completed with status ${status}`);

```

## DNS binding

| Function                                | Description                                                                           |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <code>dns(host)</code>                  | Default bundle: A, AAAA, MX, TXT, NS, CNAME, SOA. Returns a Map keyed by record type. |
| <code>dns(host, types_str)</code>       | Custom types: "A,AAAA,MX" .                                                           |
| <code>dns(host, types_str, opts)</code> | <code>opts</code> may contain <code>servers: "1.1.1.1,8.8.8.8"</code> .               |

### Examples

```
let r = dns("example.com");
print(`A: ${r.A.join(", ")}`);
print(`MX: ${r.MX.join(", ")}`);

// Custom resolver
let r = dns("example.com", "A,AAAA", #{ servers: "127.0.0.1:5353" });
print(r);

// Check DNSSEC chain presence
let r = dns("example.com", "DNSKEY,DS");
if r.DNSKEY.len() == 0 { print("no DNSKEY"); }
```

## TLS probe binding

| Function                        | Description                                                                                                                                                                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>tls(host [, opts])</code> | Handshake + cert introspection. Returns a Map with <code>subject</code> , <code>issuer</code> , <code>sans</code> , <code>not_before</code> , <code>not_after</code> , <code>days_remaining</code> , <code>fingerprints</code> , <code>chain</code> , <code>tls_version</code> , <code>cipher_suite</code> . |

### Examples

```

let t = tls("example.com:443");
print(`CN: ${t.subject}`);
print(`issuer: ${t.issuer}`);
print(`expires in ${t.days_remaining} days`);
if t.days_remaining < 14 { eprint("cert expiring soon"); return 1; }

// Verify an expected subject
let t = tls("api.example.com:443");
assert(t.sans.contains("api.example.com"), "SAN mismatch");

// Per-host SNI override
let t = tls("alt.example.com:443", #{ sni: "primary.example.com" });

```

## Ping binding

| Function                      | Description                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>ping(host)</code>       | Default TCP ping, 4 packets. Returns <code>#{ sent, received, loss_pct, min_ms, avg_ms, max_ms, ...}</code> . |
| <code>ping(host, opts)</code> | <code>opts</code> supports <code>count</code> , <code>icmp: true</code> , <code>timeout_ms</code> .           |

## Examples

```

let r = ping("8.8.8.8");
print(`${r.received}/${r.sent} received, avg ${r.avg_ms}ms`);

// ICMP (needs privileges on macOS/Linux for non-DGRAM types)
let r = ping("example.com", #{ icmp: true, count: 10 });

// TCP ping to a specific port
let r = ping("example.com:443");
assert(r.received > 0, "no reachability");

```

## File transfer bindings

### FTP

| Function                    | Description                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------|
| <code>ftp(url)</code>       | FTP / FTPS anonymous listing or retrieve.                                                |
| <code>ftp(url, opts)</code> | <code>opts</code> : <code>user</code> , <code>pass</code> , <code>ftps_implicit</code> . |

## SFTP

| Function                     | Description                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>sftp(url)</code>       | SSH-backed listing or retrieve.                                                                        |
| <code>sftp(url, opts)</code> | <code>opts</code> : <code>user</code> , <code>pass</code> , <code>privkey</code> , <code>port</code> . |

## TFTP

| Function                     | Description                                          |
|------------------------------|------------------------------------------------------|
| <code>tftp(url)</code>       | RFC 1350 download.                                   |
| <code>tftp(url, opts)</code> | <code>opts</code> : <code>blksize</code> (RFC 2348). |

## Gopher

| Function                 | Description              |
|--------------------------|--------------------------|
| <code>gopher(url)</code> | RFC 1436 selector fetch. |

## Examples

```
let listing = ftp("ftp://ftp.example.com/");
print(`${listing.entries.len()} entries`);

let data = ftp("ftp://ftp.example.com/file.bin");
print(`${data.size} bytes`);

let h = sftp("sftp://me@example.com/srv/app.log", #{ privkey:
"~/.ssh/ci_rsa" });
file_write_all("/tmp/app.log", h.body);

let img = tftp("tftp://tftp.example.com/boot.img", #{ blksize: 8192 });
print(img.size);

let g = gopher("gopher://gopher.floodgap.com/0/gopher/proxy");
print_raw(g.body);
```

## Mail retrieval bindings

---

### POP3

| Function                     | Description                                                                         |
|------------------------------|-------------------------------------------------------------------------------------|
| <code>pop3(url)</code>       | Capability + message listing.                                                       |
| <code>pop3(url, opts)</code> | <code>opts</code> : <code>stls</code> , <code>retrieve_index</code> (int, 1-based). |

## IMAP

| Function                     | Description                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------|
| <code>imap(url)</code>       | EXAMINE + capabilities.                                                                               |
| <code>imap(url, opts)</code> | <code>opts</code> : <code>fetch</code> (int — message index), <code>peek</code> : <code>true</code> . |

## Examples

```
// POP3 probe with STLS
let r = pop3("pop3://alice:s3cr3t@pop.example.com/", #{ stls: true });
print(`${r.message_count} messages`);

let first = pop3s("pop3s://alice:s3cr3t@pop.example.com/", #{
  retrieve_index: 1 });
print_raw(first.body);

// IMAP peek
let r = imap("imaps://alice:s3cr3t@imap.example.com/INBOX", #{ fetch: 3,
  peek: true });
print_raw(r.body);
```

## SMTP binding

| Function                     | Description                                                                                                                                                                                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>smtp(url)</code>       | Capability + STARTTLS probe.                                                                                                                                                                                                                                                                                           |
| <code>smtp(url, opts)</code> | Send a message. <code>opts</code> keys: <code>from</code> , <code>to</code> (array), <code>subject</code> , <code>body</code> , <code>headers</code> , <code>auth</code> ("user:pass"), <code>helo</code> , <code>no_starttls</code> , <code>dkim_key</code> , <code>dkim_selector</code> , <code>dkim_domain</code> . |

## Examples

```
// Probe only
let r = smtp("smtp://mail.example.com:25");
print(`capabilities: ${r.capabilities.join(", ")}`);

// Send
smtp("smtp://smtp.example.com:587", #{
  from: "me@example.com",
  to: ["you@example.com"],
  subject: "Automated notice",
  body: "Probe result: OK",
  auth: env("SMTP_CREDS"),
});

// DKIM-signed
smtp("smtp://smtp.example.com:587", #{
  from: "me@example.com",
  to: ["you@example.com"],
  subject: "Signed",
  body: "This message is DKIM-signed.",
  dkim_key: "~/dkim/default.pem",
  dkim_selector: "default",
});
```

## WebSocket binding

| Function                   | Description                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------|
| <code>ws(url)</code>       | Handshake + Ping/Pong. Returns <code>#{ status, duration_ms, negotiated_protocol }</code> . |
| <code>ws(url, opts)</code> | <code>opts: headers, subprotocols, send_text, send_binary.</code>                           |

## Examples

```
let r = ws("wss://echo.websocket.events");
print(`connected in ${r.duration_ms}ms`);

// Send a frame, check response
let r = ws("wss://echo.websocket.events", #{ send_text: "hello" });
print(r.received);
```

## Other protocol bindings

Brief interface reference; each has at least one example in `script/*.rhai`.

## LDAP

```
let r = ldap("ldap://ldap.example.com");           // anonymous RootDSE query
```

## RTSP

```
let r = rtsp("rtsp://camera.example.com/stream1");
```

## Dict

```
let r = dict("dict://dict.org/d:recon");
```

## NTP

```
let r = ntp("ntp://pool.ntp.org");  
print(`offset: ${r.offset_ms}ms, rtt: ${r.rtt_ms}ms`);
```

## Memcached

```
let r = memcached("memcache://cache.example.com/?version");  
let s = memcached("memcache://cache.example.com/", #{ command: "stats" });
```

## Redis

```
let r = redis("redis://cache.example.com/");       // PING  
let info = redis("redis://cache.example.com/", #{ command: "INFO server"  
});
```

## MQTT

```
let r = mqtt("mqtt://broker.example.com", #{  
    topic: "sensors/room-1",  
    message: '{"temp": 21.3}',  
});
```

## Encoding & decoding bindings

---

0.55.0 expanded encode with Aztec + PDF417 and added decode.

| Function                                                       | Description                                                                                     |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>encode::qr(text)</code>                                  | PNG Blob.                                                                                       |
| <code>encode::datamatrix(text)</code>                          | PNG Blob.                                                                                       |
| <code>encode::barcode(format, text)</code>                     | PNG Blob for any 1D/2D.                                                                         |
| <code>encode::encode(format, text)</code>                      | PNG Blob (default renderer).                                                                    |
| <code>encode::encode(format, text, output_format)</code>       | <code>output_format = ascii / svg / png</code> .<br>Returns string for ascii/svg, Blob for png. |
| <code>encode::encode(format, text, output_format, opts)</code> | <code>opts: qr_level: "L" "M" "Q" "H"</code> .                                                  |
| <code>encode::decode(blob)</code>                              | Scan an image Blob → <code>{ text, format }</code> .                                            |
| <code>encode::list()</code>                                    | Supported format names.                                                                         |

## Examples

```
// Generate QR
let png = encode::qr("https://example.com");
file_write_all("/tmp/site.png", png);

// Tune QR durability
let png = encode::encode("qr", "sticker-text", "png", #{ qr_level: "H" });

// Get SVG
let svg = encode::encode("qr", "recon", "svg");
file_write_all("/tmp/site.svg", svg);

// Decode
let data = file_read("/tmp/site.png");
let r = encode::decode(data);
print(`${r.format}: ${r.text}`);

// Round-trip test
let png = encode::qr("hello");
let r = encode::decode(png);
assert(r.text == "hello", "round-trip failed");
```

## Hashing binding

Ten algorithms plus CRC32.

| Function                                                                              | Description           |
|---------------------------------------------------------------------------------------|-----------------------|
| <code>md5(data)</code>                                                                | MD5 (hex string).     |
| <code>sha1(data)</code>                                                               | SHA-1.                |
| <code>sha224</code> / <code>sha256</code> / <code>sha384</code> / <code>sha512</code> | SHA-2 family.         |
| <code>sha3_256</code> / <code>sha3_512</code>                                         | SHA-3.                |
| <code>blake3(data)</code>                                                             | BLAKE3.               |
| <code>crc32(data)</code>                                                              | CRC32 (hex, 8 chars). |
| <code>hmac_sha256(key, data)</code>                                                   | Keyed HMAC.           |

`data` accepts string or Blob.

## Examples

```
let h = sha256(file_read("big.iso"));
print(h);

// HMAC
let sig = hmac_sha256("sharedsecret", "payload");
print(sig);

// Hash an HTTP body
let r = http("https://example.com/");
let fp = blake3(r.body);
print(`fingerprint: ${fp}`);
```

## Compression & archive bindings

### Compression

| Function                                                 | Description                                                                                                                                                                            |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>compression::compress(algo, data [, level])</code> | <code>algo</code> = <code>gzip</code> , <code>deflate</code> , <code>brotli</code> , <code>zstd</code> , <code>bzip2</code> , <code>lz4</code> , <code>xz</code> , <code>snap</code> . |
| <code>compression::decompress(algo, data)</code>         | Reverse.                                                                                                                                                                               |
| <code>compression::decompress_auto(data)</code>          | Auto-detect from magic bytes.                                                                                                                                                          |
| <code>compression::list()</code>                         | Supported algos.                                                                                                                                                                       |

### Archive

| Function                                               | Description                            |
|--------------------------------------------------------|----------------------------------------|
| <code>archive::create(dest_path, sources_array)</code> | Infer format from extension.           |
| <code>archive::extract(src_path, dest_dir)</code>      | Extract zip/tar/tar.gz/tar.xz/tar.bz2. |

## Examples

```
// Round-trip
let body = http("https://example.com/page.html").body;
let gz = compression::compress("gzip", body, 6);
print(`${body.len()} → ${gz.len()} bytes`);
let back = compression::decompress("gzip", gz);
assert(back.len() == body.len(), "round-trip");

// Zip up
archive::create("/tmp/backup.zip", ["file1.txt", "dir1/"]);
archive::extract("/tmp/backup.zip", "/tmp/restore/");
```

## Encryption bindings

age + PGP shell-out.

| Function                                                   | Description                                                         |
|------------------------------------------------------------|---------------------------------------------------------------------|
| <code>encrypt::keygen()</code>                             | New X25519 age keypair. Returns <code>#{ public, private }</code> . |
| <code>encrypt::encrypt(data, recipients)</code>            | <code>recipients</code> = array of age-pub strings.                 |
| <code>encrypt::decrypt(data, identity)</code>              | <code>identity</code> = secret key string.                          |
| <code>encrypt::encrypt_passphrase(data, passphrase)</code> | Script-based.                                                       |
| <code>encrypt::decrypt_passphrase(data, passphrase)</code> | —                                                                   |

## Examples

```

let kp = encrypt::keygen();
print(`public: ${kp.public}`);
file_write_all("~/age/identity", kp.private);

let plaintext = "highly classified";
let ct = encrypt::encrypt(plaintext, [kp.public]);
let pt = encrypt::decrypt(ct, kp.private);
assert(pt.to_string() == plaintext, "round-trip");

let ct2 = encrypt::encrypt_passphrase("hello", "correct horse battery
staple");

```

## Check-digit binding

| Function                                       | Description              |
|------------------------------------------------|--------------------------|
| <code>checkdigit::verify(algo, input)</code>   | true / false.            |
| <code>checkdigit::create(algo, partial)</code> | Computed full value.     |
| <code>checkdigit::list()</code>                | All 60+ algorithm names. |

### Examples

```

assert(checkdigit::verify("isbn13", "9780131103627"), "valid ISBN");

let full = checkdigit::create("ean13", "400638133393");
print(full);           // 4006381333931

for algo in checkdigit::list() {
    print(algo);
}

```

## Sample-data binding

| Function                       | Description                       |
|--------------------------------|-----------------------------------|
| <code>sample::get(name)</code> | String content of a named sample. |
| <code>sample::list()</code>    | Array of sample names.            |

### Examples

```
let u = sample::get("json-user");
print(u);

for s in sample::list() { print(s); }
```

## JWT binding

| Function                                  | Description                                  |
|-------------------------------------------|----------------------------------------------|
| <code>jwt::view(token)</code>             | Decode header + claims (no signature check). |
| <code>jwt::sign(payload_map, opts)</code> | opts: alg, secret or key path.               |
| <code>jwt::validate(token, opts)</code>   | opts: alg, secret or key / pubkey.           |

### Examples

```
let t = jwt::sign({ sub: "alice", exp: now() + 3600 }, {
  alg: "HS256",
  secret: env("JWT_SECRET"),
});

let ok = jwt::validate(t, { alg: "HS256", secret: env("JWT_SECRET") });
assert(ok, "validation failed");

let parts = jwt::view(t);
print(`alg: ${parts.header.alg}, sub: ${parts.payload.sub}`);
```

## Email-protection binding

| Function                                      | Description               |
|-----------------------------------------------|---------------------------|
| <code>email::spf(domain)</code>               | SPF record + validation.  |
| <code>email::dmarc(domain)</code>             | DMARC policy.             |
| <code>email::dkim(domain, selector)</code>    | DKIM record for selector. |
| <code>email::mta_sts(domain)</code>           | MTA-STS policy.           |
| <code>email::bimi(domain [, selector])</code> | BIMI record.              |
| <code>email::tls_rpt(domain)</code>           | SMTP TLS-RPT record.      |

### Examples

```

let s = email::spf("example.com");
if s.valid { print("SPF OK"); } else { print(`SPF: ${s.error}`); }

let d = email::dmarc("example.com");
print(`policy: ${d.policy}`);

let dk = email::dkim("example.com", "selector1");
print(dk.record);

```

## Netstatus binding

| Function                      | Description                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------|
| <code>netstatus::run()</code> | Execute the probe bundle defined in <code>~/.recon/config.toml</code> [netstatus] . |

```

let r = netstatus::run();
for p in r.probes {
    print(`${p.name}: ${p.status}`);
}

```

## Text-encoding binding

| Function                                        | Description                                                           |
|-------------------------------------------------|-----------------------------------------------------------------------|
| <code>text::decode(blob, charset)</code>        | Decode bytes → string.                                                |
| <code>text::encode(string, charset)</code>      | Encode string → bytes.                                                |
| <code>text::detect(blob)</code>                 | Auto-detect ( <code>chardetng</code> ). Returns charset label.        |
| <code>text::normalize_newlines(s, style)</code> | style = <code>unix</code> / <code>windows</code> / <code>mac</code> . |
| <code>text::list_charsets()</code>              | Supported labels.                                                     |

## Examples

```

let bytes = file_read("greek.txt");
let charset = text::detect(bytes);
let utf8 = text::decode(bytes, charset);
file_write_all("greek-utf8.txt", text::encode(utf8, "utf-8"));

let crlf = text::normalize_newlines("line1\nline2\n", "windows");

```

## Whois binding

| Function                   | Description                                                |
|----------------------------|------------------------------------------------------------|
| <code>whois(domain)</code> | Two-hop whois (registry then registrar). Returns raw text. |

```
let w = whois("example.com");
print_raw(w);
```

## IPFS binding

| Function               | Description                                                                    |
|------------------------|--------------------------------------------------------------------------------|
| <code>ipfs(url)</code> | Fetch <code>ipfs://</code> or <code>ipns://</code> via the configured gateway. |

```
let data = ipfs("ipfs://QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG");
print(data.body.len());
```

## Agent-browser binding

Exposed as a static module. Requires `agent-browser` on `$PATH`.

| Function                                        | Description                                  |
|-------------------------------------------------|----------------------------------------------|
| <code>agentBrowser::available</code>            | Bool.                                        |
| <code>agentBrowser::version</code>              | Version string.                              |
| <code>agentBrowser::open(url)</code>            | Navigate to URL.                             |
| <code>agentBrowser::close()</code>              | End the session.                             |
| <code>agentBrowser::click(selector)</code>      | Click by CSS selector or <code>@ref</code> . |
| <code>agentBrowser::fill(selector, text)</code> | Fill a form field.                           |
| <code>agentBrowser::type(selector, text)</code> | Type into a field (character events).        |
| <code>agentBrowser::keyboard_type(text)</code>  | Keyboard-level typing.                       |
| <code>agentBrowser::press(key)</code>           | Key press (e.g. "Enter", "Tab").             |
| <code>agentBrowser::screenshot(path)</code>     | Save PNG.                                    |
| <code>agentBrowser::pdf(path)</code>            | Save PDF.                                    |
| <code>agentBrowser::snapshot()</code>           | Accessibility-tree dump.                     |
| <code>agentBrowser::eval(js)</code>             | Execute JS, return the value.                |

| Function                                   | Description                 |
|--------------------------------------------|-----------------------------|
| <code>agentBrowser::get(selector)</code>   | Read innerText / value.     |
| <code>agentBrowser::find(selector)</code>  | Locate (returns ref).       |
| <code>agentBrowser::cmd(args_array)</code> | Raw passthrough to the CLI. |

## Examples

```
if !agentBrowser::available { return 2; }

agentBrowser::open("https://example.com/login");
agentBrowser::fill("#user", "alice");
agentBrowser::fill("#pass", "s3cr3t");
agentBrowser::click("button[type=submit]");
agentBrowser::screenshot("/tmp/after-login.png");
agentBrowser::close();
```

## SQLite binding

| Function                                       | Description                                                                                                                      |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <code>sqlite(spec) / sqlite(spec, mode)</code> | Open. spec = path, <code>":memory:"</code> , or an alias. mode = <code>ro</code>   <code>rw</code> (default)   <code>rw</code> . |
| <code>.query(sql) / .query(sql, params)</code> | Array of Maps.                                                                                                                   |
| <code>.query_one(sql, params)</code>           | First row as a Map (or <code>()</code> ).                                                                                        |
| <code>.query_value(sql, params)</code>         | Single scalar value.                                                                                                             |
| <code>.exec(sql, params)</code>                | Row count affected.                                                                                                              |

## Examples

```

let db = sqlite(":memory:");
db.exec("CREATE TABLE users (id INTEGER PRIMARY KEY, name TEXT);");
db.exec("INSERT INTO users (name) VALUES (?)", ["alice"]);
db.exec("INSERT INTO users (name) VALUES (?)", ["bob"]);

let rows = db.query("SELECT id, name FROM users ORDER BY id");
for r in rows {
    print(`${r.id}: ${r.name}`);
}

let count = db.query_value("SELECT COUNT(*) FROM users");
print(`total: ${count}`);

```

## Document-conversion bindings

0.58.0. comrak for HTML; agent-browser for PDF.

| Function                                                                  | Description                                                |
|---------------------------------------------------------------------------|------------------------------------------------------------|
| <code>md_to_html(md)</code> / <code>md_to_html(md, opts)</code>           | Markdown (string or Blob) → HTML string.                   |
| <code>md_to_pdf(md, dest)</code> / <code>md_to_pdf(md, dest, opts)</code> | Markdown → PDF at <code>dest</code> . Needs agent-browser. |
| <code>html_to_pdf(html, dest)</code>                                      | HTML → PDF at <code>dest</code> . Needs agent-browser.     |

### Opts map

| Key                         | Default          | Description                                |
|-----------------------------|------------------|--------------------------------------------|
| <code>toc</code>            | false            | Inject linkable TOC.                       |
| <code>toc_depth</code>      | 3                | Include H1..H N.                           |
| <code>toc_title</code>      | "Contents"       | —                                          |
| <code>title</code>          | document default | <code>&lt;title&gt;</code> + PDF metadata. |
| <code>css</code>            | —                | Additional CSS (appended after default).   |
| <code>no_default_css</code> | false            | Skip bundled default CSS.                  |
| <code>gfm</code>            | false            | GitHub-flavored extensions.                |

### Examples

```
let md = file_read("README.md").to_string();

// md → html
let html = md_to_html(md, #{ toc: true, toc_depth: 3, gfm: true, title:
"README" });
file_write_all("/tmp/readme.html", html);

// md → pdf
md_to_pdf(md, "/tmp/readme.pdf", #{ toc: true, gfm: true, title: "README"
});

// html → pdf
let html = http("https://example.com/report.html").body.to_string();
html_to_pdf(html, "/tmp/report.pdf");

// Fully custom styling
md_to_pdf(md, "/tmp/styled.pdf", #{
  no_default_css: true,
  css: file_read("print.css").to_string(),
  toc: true,
  toc_title: "Table of Contents",
});
```

---

# Part IV — Appendices

---

## Exit codes

---

Curl-compatible exit codes for common cases:

| Code | Meaning                                                                   |
|------|---------------------------------------------------------------------------|
| 0    | Success.                                                                  |
| 1    | Generic protocol error.                                                   |
| 2    | Source-load error ( <code>--compare</code> , etc.).                       |
| 3    | Agent-browser / Chrome unavailable (PDF features).                        |
| 6    | DNS resolution failed.                                                    |
| 7    | Connection refused.                                                       |
| 22   | HTTP $\geq 400$ with <code>-f</code> (or <code>--fail-with-body</code> ). |
| 28   | Operation timeout ( <code>--max-time</code> ).                            |
| 35   | TLS handshake error.                                                      |
| 47   | Redirect limit exceeded.                                                  |
| 52   | Empty reply from server.                                                  |
| 55   | Send error.                                                               |
| 56   | Receive error.                                                            |
| 60   | Peer cert cannot be authenticated.                                        |

Script-mode exit codes:

- `return N` from the top-level script becomes the process exit code (mod 256).
- Unhandled script errors → exit 1 (or the last `ProtocolExitCode` tag).

## Environment variables

---

| Variable                                          | Read by              | Description                                   |
|---------------------------------------------------|----------------------|-----------------------------------------------|
| <code>HTTP_PROXY</code> / <code>http_proxy</code> | <code>--proxy</code> | Proxy for <a href="http://">http://</a> URLs. |

| Variable                  | Read by             | Description                                                    |
|---------------------------|---------------------|----------------------------------------------------------------|
| HTTPS_PROXY / https_proxy | --proxy             | Proxy for <a href="https://">https://</a> URLs.                |
| ALL_PROXY / all_proxy     | --proxy             | Fallback proxy.                                                |
| NO_PROXY / no_proxy       | --noproxy           | Bypass list.                                                   |
| SSL_CERT_FILE             | --cacert            | Extra trust-root bundle.                                       |
| CURL_CA_BUNDLE            | --cacert            | Same as SSL_CERT_FILE.                                         |
| RECON_NO_PAGER            | --help / --examples | Disable paging.                                                |
| RECON_IPFS_GATEWAY        | --ipfs-gateway      | Default IPFS gateway.                                          |
| RECON_CONFIG              | config file         | Override path to config.toml .                                 |
| AGENT_BROWSER_JSON        | —                   | When 1 , agent-browser returns JSON (set internally by recon). |
| EDITOR                    | --editor            | Editor binary for response inspection.                         |
| HOME                      | various             | Used to locate ~/.recon/ .                                     |
| PAGER                     | --help / --examples | Override default less -FRX .                                   |

## Configuration file

~/.recon/config.toml — commented skeleton created by recon --init .

## Structure

```

# Default flag overrides
[defaults]
connect_timeout = 30
max_time = 60
user_agent = "recon/0.58.1 (auto)"

# Netstatus probe bundle
[netstatus]
probes = [
    { name = "dns", target = "8.8.8.8" },
    { name = "http", url = "https://example.com/" },
    { name = "tls", target = "example.com:443" },
]

# Per-host SNI → cert mapping for --serve-tls
[[serve_sni]]
host = "a.example.com"
cert = "/etc/certs/a.pem"
key = "/etc/certs/a.key"

[[serve_sni]]
host = "b.example.com"
cert = "/etc/certs/b.pem"
key = "/etc/certs/b.key"

```

## ~/.recon/ layout

Created by `recon --init`:

```

~/.recon/
├─ config.toml           # Commented skeleton; overrideable via
RECON_CONFIG
├─ script/              # Bare-word scripts (`recon --script NAME`)
│   └─ *.rhai
├─ jars/                # Persistent cookie jars
│   └─ <session>.db
├─ sni/                 # Per-host TLS cert + key pairs for --serve-
tls
│   └─ a.example.com/cert.pem
│   └─ a.example.com/key.pem
└─ hsts.txt             # Optional HSTS cache (curl-compatible)

```

## Glossary

- **Sticky session** — a `browser()` handle whose cookies + headers persist across calls. Pair with `use_persistent_session` for cross-invocation persistence.
  - **Script defaults** — the frozen snapshot of CLI flags (`-H`, `-k`, `--connect-timeout`, ...) that every script binding inherits when the caller doesn't override via an opts map.
  - **Bare-word script** — one that can be invoked by name because it lives in `~/.recon/script/`, e.g. `recon --script http` → `~/.recon/script/http.rhai`.
  - **Protocol exit code** — recon's internal scheme for passing an exit code through a panic-style error path from protocol bindings (so a failed SMTP probe can exit with curl's `CURLE_URL_MALFORMAT`, for instance).
  - **agent-browser** — external CLI that wraps Chrome DevTools Protocol. Used for `--browser-screenshot`, `--md-to-pdf`, `--html-to-pdf`, and the `agentBrowser::*` script bindings.
- 

*End of manual. For the curated-example companion, run `recon --examples`. For topic-specific deep dives, run `recon --help <topic>`.*